Supplementary Materials: Provenance-Bound Trust-Aware Ingestion for Industrial IoT Edge-to-Cloud Data Pipelines: Per-Identity Conflict Containment and Its Security–Availability Trade-offs

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This document supports the article; works are cited by author, venue and year and appear in the article's reference list. Every number is generated from the per-run raw results of the research package by its analysis scripts; Data S1 contains the per-cell summaries and all confirmatory comparisons as CSV files. The article names its evaluation stages by their purpose (main evaluation, re-execution, review-schedule evaluation, selective-hold grid, factorial cells and controls, sensitivity analyses, public-telemetry replay, lifecycle analysis). Repository file and directory names keep the internal identifiers under which the stages were run (for example `A1_review`, `research/V4`), because the recorded hashes refer to them; Table S8 maps each stage to its identifiers. Pipeline subscripts: $B5_{60/10}$, one-shot review rule with minimum case age $R = 60$ s and cadence $C = 10$ s; $B5_{DR/C}$, deferring rule with R and C in seconds; $B5_{0/10}$, one-shot rule at $R = 0$ s, $C = 10$ s; $B5_{ev}$, event-driven review; likewise for PBTI-R.

S1. Positioning

Table S1. Coverage of mechanism components by the 15 works examined first in the closest-prior-work search, most of them IoT-specific (Y: in the design; Y*: designed and evaluated for that property; P: partial; D: discussed only; –: not reported in the examined text; n/e: abstract only). C1 per-event device signature; C2 gateway signing for legacy devices; C3 workload identity / countersigned relay statement; C4 path-bound admission policy; C5 credential status; C6 replay versus retry; C7 duplicate/equivocation detection; C8 ordering/late data; C9 disconnection/backlog recovery; C10 chained or Merkle evidence; C11 content-addressed raw storage; C12 lineage and taint; C13 industrial evaluation; C14 per-item trust scoring. For PBTI, C6 is P because the replay/retry label is assigned from context and does not affect authority (Section 4.2 of the article). The table records which components each work reports; it is not a gap analysis. The closest analogues of the conflict response (Secure Scuttlebutt, PeerReview, the Global Data Plane, SUNDR, SCITT, the ACE profile, C2PA, Taser and Warp, Kafka and CloudEvents) are compared in Table 1 and Section 2 of the article. [a]: only the abstract was available to the reference check (Section S3), so "–" in this row means "not stated in the abstract", not an established absence. †: the full text could not be opened in the reference check, so this row could not be re-verified. ‡: its reputation scores rate the oracle nodes that supply data, combined with truth inference over the data (full text checked).

Work	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14
Buccafurri et al., MQTT-I (TDSC 2024)	P	–	–	–	–	–	P	Y	–	Y	–	–	–	–
Nawab, WedgeChain (ICDE 2021)	Y	–	P	–	–	D	Y	P	–	Y	P	–	–	–
Saito (R10-HTC 2025)	Y	–	–	–	–	–	–	–	P*	Y	–	–	–	–
Tsai et al. (Sensors 2026)	Y	–	–	Y	Y	–	–	P	–	Y	Y	P	P	–
Dahlmanns et al., ENTRUST (SaT-CPS 2021)	P	–	–	Y	P	P	Y	P	–	–	–	–	P	–
Lazzaro & Buccafurri (Future Internet 2024)	–	–	–	P	–	–	P	P	–	Y	P	–	–	–
Hasan et al. (ACM TOS 2009)	P	–	P	–	–	–	P	P	–	Y	P	P	–	–
Aman et al. (IoTPTS 2017; IEEE IoT-J 2019)[a]	P	P	–	P	P	P	–	–	–	–	–	–	–	–
Javaid et al., BlockPro (BlockSys 2018)[a]	P	–	–	P	P	P	–	–	–	Y	P	–	–	–
Lim et al. (DMSN 2010)[a]	–	–	–	–	–	–	–	–	–	–	–	–	–	Y*
Rizzardi et al., AUPS (Inf. Syst. 2016)†	–	–	–	Y	P	D	–	–	–	–	–	–	–	P
Gigli et al., ZONIA (IEEE IoT-J 2025)‡	–	–	–	P	–	–	–	–	–	P	–	–	–	Y*
James et al., MQTT-ZT (CSR 2025)	n/e	n/e	n/e	Y	n/e	n/e	n/e	n/e	n/e	n/e	n/e	n/e	n/e	n/e
Sukumaran et al. (Information 2026)	–	–	–	–	–	–	–	–	–	Y	P	P	–	–
Matias & Macêdo (WTF 2026)	–	–	–	–	–	–	–	P	Y*	–	–	–	Y	–
PBTI (this work)	Y*	Y*	Y*	Y*	Y*	P	Y*	Y*	Y*	Y*	Y*	Y*	P	–

Search coverage and what was examined. The first search used Crossref, arXiv and web search; dblp, OpenAlex and Semantic Scholar were unavailable then and IEEE Xplore and ACM Digital Library full texts could not be fetched, and a later check additionally used OpenAlex and DataCite. The table records the first search only; works on fork handling added afterwards (SUNDR, Byzantine-fault-tolerant CRDTs, the 2P-BFT-Log preprint) and recent works from a final search are discussed in the article but not scored. The automated reference check of Section S3 examined the full or standard texts of the works on which the comparisons of the article's Section 2 and Table 1 rest (Secure Scuttlebutt and its protocol guide, PeerReview, the Global Data Plane, SUNDR, the Byzantine-fault-tolerant CRDTs, the 2P-BFT-Log preprint, SCITT, Certificate Transparency, RFC 9338, the ACE publish/subscribe draft, WedgeChain, Kafka's idempotent producer and CloudEvents) for their fork evidence, post-fork response, recovery, authoritative-data semantics and trust assumptions. It had only the abstracts of Aman et al., Javaid et al., Lim et al., MQTT-ZT, SMQTT, OSCORE-IoT, GeneaLog and ProVerif, which the article cites only for what their abstracts state, only catalogue records of AUPS and of the standards of Table S43, and no text of Holm's procedure, which is cited for its standard attribution. Metadata records establish that a work exists, not the technical claim it is cited for. The absence statements describe the works and mechanisms examined and are not an exhaustive priority claim; B5 is a comparator built for this study from published fork responses, not an implementation of Secure Scuttlebutt, PeerReview or the Global Data Plane; and the 2P-BFT-Log work is a preprint (arXiv version 2; no publication record found).

S2. Protocol, State Machine and Formal Properties

Table S2. Fields of the observation statement (signed by the origin signer) and the relay statement (signed by the relaying edge workload). All fields are protected by the respective signature; odig additionally binds the observation signature.

Field	Type	Security property	Correctness property	Threat addressed
v, typ	int, string	version and statement-type domain separation	parser selection	downgrade, cross-protocol reuse
did	string	binds key subject to producer	partitions event identity	impersonation by key substitution
aid	string	device→asset authorization scope	attribution of data	mis-attribution, metadata tampering
ep	integer	bounds stale-epoch injection (drain window)	distinguishes sequence resets after reboot	false conflicts, epoch replay
seq	integer	gap and replay evidence	deterministic identity (*did*, *ep*, *seq*)	duplicates, suppression, gap-fill
ts	integer (ms)	freshness bounds (skew, horizon)	event-time windowing	back/future dating (bounded)
sch	string	schema restriction per asset	decoding	schema confusion
pl	object	integrity	measurements	tampering
prev	digest or null	fork and gap-fill evidence	offline completeness audit	history rewriting
kid	string	key binding	rotation	key confusion
wid	SPIFFE ID	relay path identity reaching the verifier	path attribution	lateral movement, path spoofing
wkid	string	relay key binding	rotation	key confusion
odig	digest	binds one signed observation (content and signature)	–	re-binding of relay statements
rts	integer (ms)	evidence only	buffering statistics	–
mode	live/buffered	selects quarantine instead of rejection for a revoked key (Algorithm 1, step 5); otherwise evidence only	RECOVERED annotation	–

Canonical form: RFC 8785 JCS (ASCII keys, IEEE-754 numbers serialized per ECMAScript, integers within $\pm 2^{53}$); signatures: Ed25519 over a domain-separation prefix and the canonical bytes; content digest cd = SHA-256 of the canonical observation statement. Default parameters: skew 5 s; horizon $h_{\max}$ 1 h (120 s in E11b and the mixed workload); retry window 30 s; rotation grace 1 h; maximum sequence jump 100,000; epoch drain and gap closure 1 h; checkpoints every 512 records or 5 s.

Figure S1 shows how one data object is bound to its evidence and the event-identity slot machine whose transitions Table S3 lists; the article defines the Rule L classes in Section 4.2.

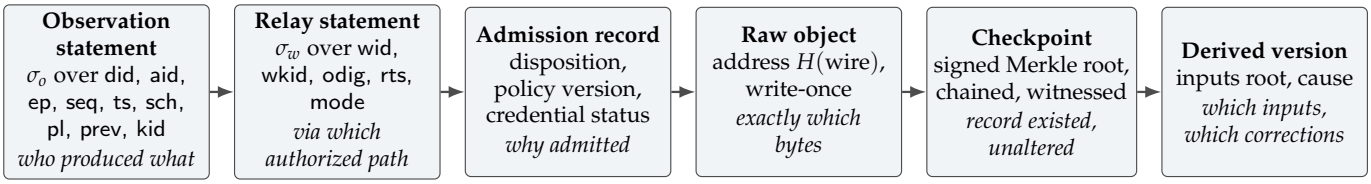
Table S3. Transitions of the event-identity slot machine (Current state + Evidence + Policy condition → Next state). Ordinary arrival transitions are distinguished from authorized review-release transitions (marked R). The default one-hour gap-closure horizon was not reached in the scenario campaigns; a regression test with a five-second horizon exercises LOST, STALE_SLOT quarantine and reviewed release, and the bounded lifecycle check (Table S5) combines gap closure, release, revocation and rotation.

Current	Evidence	Condition	Next (cause)
ABSENT	first version cd; no implied digest or implied $= cd$	admission checks passed	COMMITTED (ACCEPT)
ABSENT	first version cd; implied digest $\neq cd$	admission checks passed	CONTESTED (LINK_MISMATCH)
ABSENT	implied digest only (successor received)	–	ABSENT, expected digest known
COMMITTED	second version $cd' \neq cd$	admission checks passed	CONTESTED (EQUIVOCATION)
COMMITTED	successor with prev $\neq cd$	successor admitted	CONTESTED (LINK_MISMATCH)
COMMITTED	origin key revoked with $t_{comp} \leq t_{commit}$ (also after a revised record with an earlier t_{comp})	revocation observed by verifier; for a slot released under the revoked key: its release condition fails under the current record	COMMITTED+TAINTED (REVOCATION)
ABSENT in open gap	gap age > gap-closure horizon	periodic tick	LOST (GAP_CLOSED)
CONTESTED	any further version	–	CONTESTED (absorbing; evidence kept)
LOST	arriving version without an authorized waiver	ordinary admission	LOST (receipt QUARANTINE STALE_SLOT)
LOST (R)	version re-evaluated through an authorized release	STALE_SLOT is the only waived reason; all other admission checks pass	COMMITTED or CONTESTED under Rule L; on failure no authoritative observation (RELEASE_FAILED)
ABSENT (R)	quarantined version (STALE_TS, STALE_EPOCH, REVOKED_BUFFERED, ...) re-evaluated through an authorized release	only the case's reason waived; REVOKED_BUFFERED also requires the release-evidence condition (discretionary: $ts < t_{comp}$; witness: token for the exact digest before t_{low})	COMMITTED or CONTESTED under Rule L; a conflict records the version as evidence, nothing becomes authoritative
any	identical cd already admitted	–	unchanged (DUPLICATE RETRY or SUSPECTED_REPLAY)
any	reject or quarantine condition (Algorithm 1)	–	unchanged (REJECT / QUARANTINE)

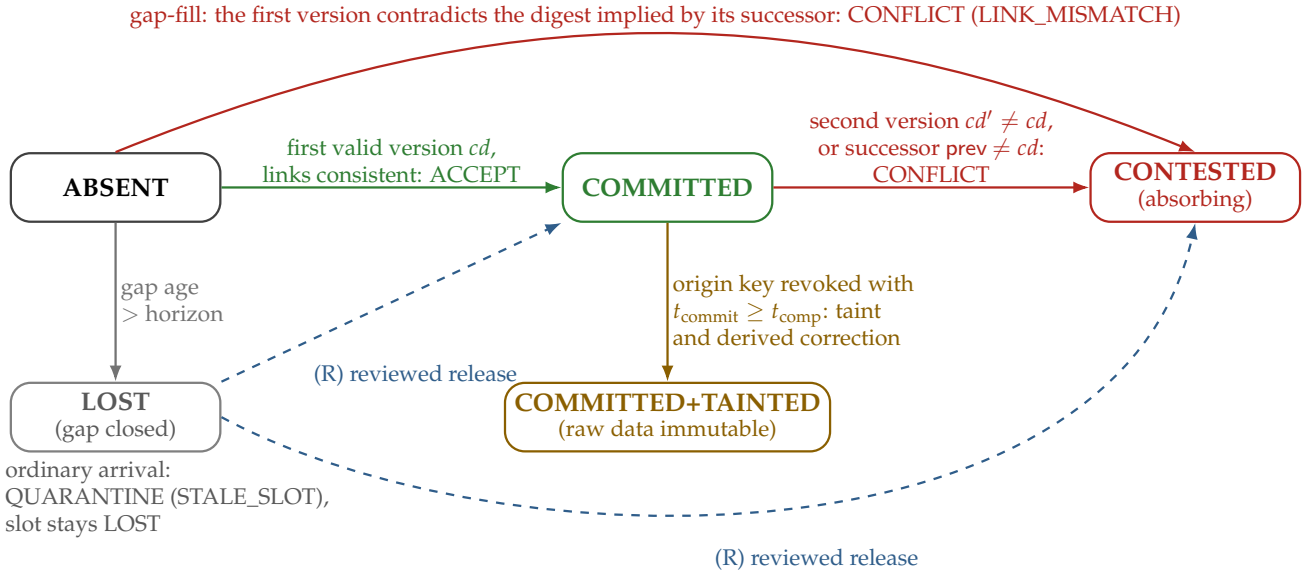
Table S4. Invariants, their oracles and the scenarios that exercise them.

Invariant	**Statement**	**Oracle (scenarios)**
I1 Integrity	modifying any protected field without the key yields REJECT and no state change	unit test over all 18 protected fields; non-legitimate admissions (E2, E3, E6)
I2 Authorization	every committed slot has an ACCEPT record with PERMIT under the policy in force	registry audit; unauthorized-path admissions (E4, E15)
I3 Idempotency	at most one authoritative observation per identity	duplicate authoritative count (E7, E11)
I4 Non-overwrite	two validly signed contents for one identity are never authoritative	equivocation/gap-fill admissions (E5, E8)
I5 Provenance	latest derived versions reference only authoritative inputs; roots and aggregates recompute	lineage audit (all runs)
I6 Immutability	committed raw bytes and checkpointed evidence cannot change undetected	raw-store and evidence audits (E16)
I7 Credential enforcement and taint	ordinary admission after an observed revocation is refused or quarantined; an explicitly authorized release exception is audited separately; the selector taints origin-key commits at or after t_{comp} (commit time, not proof of creation time; re-run on revised records)	E14 ordinary-admission and taint counts; release-then-taint regressions and the lifecycle analysis (Table S29)
I8 Recovery	buffered observations within horizon committed exactly once (a suspected-replay label may occur but is non-authoritative)	missing and duplicate counts (E11)
I9 Convergence	for a fixed admitted evidence set, final state independent of delivery order	permutation unit test; exact windows (E9, E10)
I10 Determinism	replaying the receipt log reproduces every disposition and the registry	decision-replay audit (all runs)

(a) Trust chain of one data object (what each link lets an auditor verify)



(b) Slot transitions of one event identity $e = (did, ep, seq)$ under Rule L



(R), dashed: authorized reviewed release from LOST. STALE_SLOT is the only waived reason and every other admission check is re-run; Rule L then yields COMMITTED, or CONTESTED if a contradicting digest is known. A release that fails a check leaves the slot LOST, and no failed release makes an observation authoritative.

Receipts that never change slot state
DUPLICATE (retry or suspected replay): byte-identical content; retry and replay are told apart only by delivery context.
REJECT: malformed, unknown key, signature or binding failure, revoked key, unauthorized path, asset, schema or origin.
QUARANTINE (archived, not released): stale or future time stamp, stale epoch or slot, sequence jump, key expired beyond grace, revoked key on buffered data.

Figure S1. (**a**) Trust chain of one data object and what each link lets an auditor verify: observation statement and relay statement (bound through odig), admission record, write-once raw object, signed and witnessed Merkle checkpoint, and derived version with its inputs and cause. (**b**) Event-identity slot transitions under Rule L: the state of a slot is a function of the admitted evidence set (ABSENT, COMMITTED, CONTESTED, LOST, and COMMITTED+TAINTED after revocation with a compromise time). Ordinary arrival at a LOST slot remains quarantined (STALE_SLOT); an authorized reviewed release (dashed, R) may re-run admission with STALE_SLOT as the only waived reason and reach COMMITTED or CONTESTED under Rule L, and a failed release creates no authoritative observation. The taint transition is subject to the reviewed-release conditions of Section 4.3 of the article (a slot released under the revoked key is exempt while its release condition holds under the current revocation record). Receipts that do not change slot state are listed below; Table S3 specifies the complete transitions.

Locality of Rule L: statement, proof, exclusions and liveness. The model has identities $e = (d, p, q)$ with predecessor $\text{pred}(e) = (d, p, q-1)$ for $q > 0$ and none for $q = 0$; an admitted statement s has identity $e(s)$, content digest $cd(s)$ and link $\text{prev}(s)$, a digest or $\perp$. The evidence state is $E = (V, I)$, and admission updates $V(e(s)) \leftarrow V(e(s)) \cup \{cd(s)\}$ and, if $q(e(s)) > 0$ and $\text{prev}(s) \neq \perp$, $I(\text{pred}(e(s))) \leftarrow I(\text{pred}(e(s))) \cup \{\text{prev}(s)\}$; in the implementation these are the only writes to V and I (`PBTIVerifier._history` and the `_history` methods of the historical and the corrected B5 subclass, which perform the same

two writes). An *admission* is a receipt that reaches the write to V: for PBTI these are the ACCEPT and CONFLICT receipts and the re-run admission of a reviewed release; for B5 they also include the FEED_FORK_HOLD receipts, which write V and I before the hold. Classes: ABSENT if $V(e) = \emptyset$; otherwise COMMITTED if $|V(e) \cup I(e)| = 1$ and CONTESTED if $|V(e) \cup I(e)| \geq 2$; LOST if the gap-closure flag $\lambda(e)$ is set; authority requires COMMITTED and no taint flag $\tau(e)$, and for B5 additionally that e is not held. *Lemma*: for any E and any sequence S of k admitted statements with no lifecycle transition in between, E' differs from E only on $A(S) = \{e(s_i)\} \cup \{\text{pred}(e(s_i)) : q(e(s_i)) > 0, \text{prev}(s_i) \neq \perp\}$, and $|A(S)| \leq 2k$, with equality only when the $2k$ identities are pairwise distinct. *Proof sketch*: by induction on k, since each update writes V at one identity and I at most at one other; the class of $e \notin A(S)$ is a function of $V(e)$, $I(e)$ and $\lambda(e)$ and its authority in addition of $\tau(e)$, and λ and τ are written only by gap closure and revocation processing, which are excluded. The number of identities whose class changes can reach $2k$ but is not implied by pairwise distinct identities alone. *Exclusions*: (E1) admission decisions that earlier admissions or time change (epochs, gaps, sequence jumps, horizons, key validity, revocation, policy versions); (E2) lifecycle transitions (gap closure marks a range LOST; revocation taints every qualifying commit of a key); (E3) downstream dependency fan-out; (E4) undetected contamination (a forged version for which no contradiction is admitted stays COMMITTED); (E5) the selective-hold extension, whose authority depends on feed-level suspicion; (E6) replicated verifiers. *Liveness, separate from this safety property*: (L1) a gap-fill or equivocating version is contested only if a contradictory version or successor link is admitted within the freshness, gap and epoch horizons; (L2) a quarantined or held observation is released only if a reviewer runs and the release evidence its policy requires is available; (L3) retroactive containment requires a published and processed revocation with t_{comp}, and its precision depends on how accurately t_{comp} is known. The locality claim is supported by the mathematical argument recorded in formal/LOCALITY_PROOF.md; the argument has not been verified in a proof assistant. A checklist for its review is in the repository (formal/PROOF_REVIEW_CHECKLIST.md). An AI-assisted review identified imprecisions in the statement and assumptions, which were revised. Separate executable checks found no counterexample within the stated bounded domains. An exhaustive checker written from the statement alone, without the verifier code, found no violation in 4.07×10^9 (state, batch) cases of the pure model and in 1,000 random histories (10,803 signed receipts) on the real verifier.

The lifecycle check found one defect before the confirmatory runs: after a revocation record was revised with an earlier compromise time, the verifier did not re-run the taint selector, so an observation committed after the revised compromise time stayed authoritative (minimal schedule: commit at 2 s; revoke with $t_{\text{comp}} = 3$ s; revise to $t_{\text{comp}} = 0$). The fix keeps a key revoked from its first revocation, lets a revision only move t_{comp} earlier and re-runs the selector; a record released under a revoked key stays exempt only while its release condition holds. The counterexample is kept in the repository. No scenario revokes a key twice, and the re-executed main evaluation was identical on the corrected source. The symbolic model (ProVerif 2.05; Dolev–Yao attacker, unbounded sessions, adaptive key compromise) establishes the signature-binding correspondences stated in the article; its variant with an attacker-writable device-to-gateway link shows that the device key alone then suffices. Rego parity: 34,749 of 34,749 well-formed and 115 of 115 malformed requests gave the same decision and reason in the Python decision point and in the Rego policy evaluated by rego-cpp; OPA itself was not run.

Table S5. Bounded checks of Rule L and of the lifecycle (repository `results/v4/formal/v4_locality_lifecycle_check.json`; all pass). These are bounded results, not proofs. An earlier bounded check examined Rule L exhaustively, for one device, one epoch, sequence numbers 0–3 and up to two contents per slot, on the unmodified verifier with real signatures (all 390,024 delivery orders of the admitted sets) and on a re-implementation (1,060,190 set–order pairs): the final state never depended on delivery order, contest was never undone, every equivocation and every gap-fill with an arriving successor ended contested, and duplicates changed nothing; larger feeds, several epochs, gap closure and release were not explored.

Part	Scope and properties	Cases
M (model)	implementation-independent model of (V, I); every base set of two universes (one epoch, sequence 0–2, two contents, prev $\in \{G, F, \perp\}$: 14 statements; two epochs, sequence 0–1: 12 statements) and every batch of one or two further statements: changed classes $\subseteq A(S)$, $\lvert A(S)\rvert \leq 2\lvert S\rvert$, both orders equal, contest absorbing	579,584
L (implementation)	unmodified verifier, real Ed25519 signatures: every base set of the two-epoch universe with every single further statement	24,576
L (sample)	seeded random batches of two or three statements over both universes	4,000
C (lifecycle)	every operation sequence of length $\leq$ 4 over 13 operations (deliveries over two epochs including a forged alternative and a rotated-key statement; gap closure; revocation with early or late t_{comp}; release with fresh or stale snapshot; rejection): evidence conformance, absorbing contest, authority and derived data, taint exactness, no ordinary admission under a revoked key, release safety, stale-snapshot refusal, per-delivery locality, rotation grace	30,940
C (sample)	seeded random sequences of length 5–8	20,000
C (both rows)	operations exercised over all lifecycle sequences: 251,106; releases 926; stale-snapshot refusals 10	

S3. Environment, Evaluation Stages, Evidence Levels and Reproduction

Re-execution checks. The re-execution of the main evaluation and the mixed workload under the extended source compared every stored run field by field: 1320 of 1320 runs were identical over 55,411 field comparisons (wall-clock and audit timings, audit outcomes reported separately, and two runner labels excluded). The second re-execution ran the published commands of the minimal reproduction kit (`tools/reexec/reproduce_subset.sh`) in a separate cloud environment, on a fresh copy of the repository, and compared every field except wall-clock timing with the stored rows (Table S11). Both re-executions used the same pipeline that produced the results, so they show determinism and portability, not independent reproduction. The kit records date, hardware, operating system, package versions, git commit, source and configuration hashes, exact commands, seeds, exit codes and test counts. In the second re-execution (2026-11-05; 2 vCPUs, Ubuntu 24.04.5 LTS, Python 3.13.18) the runner recorded the same source digest (`8dfae6decd80...`) and configuration digests as the stored confirmatory campaigns.

Author re-execution. A third re-execution was run by one of the authors (R.S.) on 7 November 2026, 10:30–14:09 UTC, on a laptop: Ubuntu 26.04.1 under Windows Subsystem for Linux 2, Intel Core i5-10310U (8 logical CPUs), 7.7 GiB, Python 3.14.4 with the pinned package versions, four parallel runs. It ran the minimal reproduction kit together with a check of the unzipped repository against its release manifest (744 files) and of the frozen files. The repository matched the manifest and was unchanged afterwards; the 32 frozen files were unchanged; the test suite gave 199 passed and 1 skipped, the 12 regression tests of the corrected comparator included; all 754 re-executed runs were identical to the stored rows over 77,881 field comparisons; the lifecycle matrix was identical in all 68 rows; and the bounded locality check passed all invariants. As expected, the run's source digest (`714b9b49...`) differs from the stored one (`8dfae6decd80...`) only by the corrected-comparator module, which these runners do not load, and the output of the bounded check differs from the stored output only in the recorded Python version and in that module's hash. The ProVerif models were not re-run. Because it repeats the same code and seeds on a second machine and a newer Python version, this re-execution confirms determinism and portability, not independent reproduction; it does not cover the corrected-comparator reruns, the attacker-model runs, the performance measurements or the statistical analyses.

Table S6. Execution environment recorded in the result manifests. Source digest: SHA-256 over the files in `src/` as recorded by each campaign; repository file names of the runs are given in `typewriter`. The Rego parity harness used regopy 1.5.2 (rego-cpp 1.5.2); the SQLite library version of the wall-clock runs was not recorded in their manifests. "Final" refers to the main-evaluation and performance campaigns; the review-schedule, selective-hold and re-execution campaigns recorded source digest `8dfae6decd80...` in their manifests (platform kernel 6.18.44, Python 3.13.18; Tables S8 and S11).

Item	Value
CPU	`Intel(R) Xeon(R) Processor @ 2.10GHz`
Logical CPUs / RAM	`2 / 7.84 GB`
Python	`3.13.18`
cryptography / NumPy / SciPy / pandas	`50.0.1 / 2.5.3 / 1.18.1 / 3.0.5`
Platform	`Linux-6.18.44-fc-v230-x86_64-with-glibc2.39 (ablations (seeds 1-10); ablations (seeds 11-20); audit; end-to-end; revocation (seeds 1-10); revocation (seeds 11-20); scenarios (seeds 1-10); scenarios (seeds 11-20); throughput; scaling)`
Platform	`Linux-6.18.44-fc-v233-x86_64-with-glibc2.39 (B4 mixed workload (seeds 1-20); B4 scenarios (seeds 1-20); v3_audit_rerun; v3_batching; v3_factorial; v3_mixed_b5 (B5 rows); v3_mixed_b5 (B5S rows); v3_mixed_b5 (PBTI-R rows); v3_mixed_b5_fixedB5; v3_mixed_b5s_fixed; v3_mqtt_e2e; v3_mqtt_e2e_skab; v3_replay_skab; v3_scenarios_b5 (B5 rows); v3_scenarios_b5 (B5S rows); v3_scenarios_b5_fixedB5; v3_scenarios_b5s_fixed; v3_scenarios_pbtir; v3_sweep)`
Source digest 0c21cbd40d64...	`v3_mixed_b5 (PBTI-R rows); v3_scenarios_pbtir`
Source digest 1fc3ae032b24...	`ablations (seeds 1-10); ablations (seeds 11-20); audit; end-to-end; revocation (seeds 1-10); revocation (seeds 11-20); scenarios (seeds 11-20); throughput; scaling`
Source digest 2cf3f17f3f6e...	`scenarios (seeds 1-10)`
Source digest 7a711eb89ec8...	`B4 mixed workload (seeds 1-20); B4 scenarios (seeds 1-20)`
Source digest c6305c988647...	`v3_factorial; v3_mixed_b5 (B5 rows); v3_mixed_b5 (B5S rows); v3_mixed_b5_fixedB5; v3_mixed_b5s_fixed; v3_scenarios_b5 (B5 rows); v3_scenarios_b5 (B5S rows); v3_scenarios_b5_fixedB5; v3_scenarios_b5s_fixed; v3_sweep`
Source digest dd8dcd908d82...	`v3_audit_rerun; v3_batching; v3_mqtt_e2e; v3_mqtt_e2e_skab; v3_replay_skab`
Regression check	`114 stored specifications (64 earlier scenario runs, 50 feed-level extension runs) re-run under the current source: 0 differences (results/summary/regression_check_v3.json)`
Broker / MQTT client	`mosquitto version 2.0.18 / paho-mqtt 2.1.0`

Internal AI-assisted checks. Separate internal AI-assisted checks, each given the specification, the repository and a narrow task but not the conclusions of the others, performed the following checks. They are not human reviews, and they share the tooling that wrote the code, so errors common to that tooling would not be caught. Because these checks shared the AI tooling used for development, common errors may remain undetected; they do not constitute external independent validation. (i) Oracle and time-dependent outcomes. Fifteen traces with AI-assisted derivation of expected outcomes drove the unmodified PBTI, B5 and PBTI-SH verifiers, and an implementation of the outcome definitions that shares no code with the repository recomputed the primary outcomes of 20 re-run campaign runs (seeds 1, 2, 21 and 22 and one factorial cell) from its own record of the authoritative set; all 420 run-outcome values agreed with the stored rows, and seven consistency identities held in all 7540 stored rows. The check found two counter defects that affect only Table S21 and no primary outcome (caption of that table), and it showed that W and NLX depend on the fixed 180 s drain: an observation that is never made available, and a forgery retained to the end, contribute until T_{end}, so differences between pipelines that differ in end-of-run loss grow with the drain. An exploratory screen with drains 60 s and 120 s shorter (linear correction of the stored rows, which is exact when all losses and releases precede the shortened end; interval-based classes only, without Wilcoxon tests or Holm correction) changed the margin class of 25 of the 222 registered W comparisons, all in the selective-hold analyses (Tables S25 and S26), and no gate outcome; in the grid, PBTI-SH's withholding was worse than PBTI's beyond the margin at 4 points with the registered drain, at 2 with a 120 s drain and at none with a 60 s drain. (ii) *Statistics.* A re-implementation of the registered procedures written from the plan text recomputed all 1182 comparisons from the raw rows; Holm-adjusted significance agreed in all of them, and the classes as described in the note of Table S10; it also identified deviations DV6

and DV7. (iii) Locality proof. An AI-assisted review of the proof against the update rule and the verifier, an implementation-independent exhaustive model checker (4 068 762 624 evidence-state and batch cases) and 1000 random histories (10 803 receipts) on the unmodified verifier with real signatures found no counterexample; the AI-assisted review led to the corrected statement of Proposition 1 (hypothesis stated on the schedule, taint and gap-closure flags, reviewer decisions excluded) and to the wording "proof sketch". (iv) Source code and the feed-level comparator. An AI-assisted review of PBTI, PBTI-R, PBTI-SH, B5, B5s, the simulator and the oracle against the specification documents, with a minimal reproduction of every suspicion and about 210 single runs of instrumented audit-only subclasses (the repository's classes were not edited), confirmed the Rule L slot states, duplicate handling, staleness checks, release waivers, reviewer checks and schedules, taint and revised revocations, the arm settings and the common random numbers (byte-identical receipt sequences across pipelines in two runs; identical per-label receipt counts in all 1300 campaign–cell–seed groups); the tracer matched the true authoritative set at 1247 mid-run checks. It found four minor defects. (a) B5 holds observations again that its reviewer had released, and can restart a fork after the feed resumed, when a late successor of an already contested slot arrives without new fork evidence; with this corrected in an audit-only subclass, B5's W fell by up to 0.26 s per observation in single mixed-workload runs with periodic review and by at most 0.003 s with event-driven review, O1 and NLX did not change in those runs, and the two matched-review W comparisons closest to the margin stayed worse beyond it (0.695 to 0.679 s and 0.699 to 0.690 s per observation); these single-run estimates of the audit are superseded by the complete reruns with the corrected implementation (*Comparator correction* below). (b) When the reviewer releases a stale case of a held feed, B5 can keep the observation held without an open case (no occurrence in 66 instrumented mixed-workload runs). (c) The hold counter of Table S21 (see its caption). (d) A raw field counting distinct held observations misses holds made on arrival; it is not used. Defects (a) and (b) can disadvantage B5; all four were subsequently corrected in a separate implementation (*Comparator correction* below). The audit also noted that B5's hold release does not re-run admission and that its resume rule keeps a forked feed held while the device produces (Section 5.2 of the article), that under message loss the missing-message runs of the sensitivity factorial are not always consecutive (identically for every pipeline), and that an audit pass shows internal consistency and replayability rather than correctness against the specification, because decision replay runs the same code. A reference model of B5, written from the specification before any B5 code was read (frozen at 23:50 UTC on 5 November 2026) and compared with the real B5 in 39,600 synthetic comparisons and 26 instrumented scenario runs, matched the implementation exactly once six choices that the specification leaves open were restated; without them it matched 23,781 comparisons. These choices cut both ways: reviewing cases in identity order favours B5 and PBTI-R equally (with creation order the one-shot reviewer would have lost 175 observations in one E11b run), whereas re-forking from slot state and the absence of review cases for contested slots cost B5 at most 0.014 s per observation in W and 0.0013 in AV5 on the real traces. The model confirmed defects (b) (no occurrence in 146 real runs) and (c) (hold counts 1–9% too high with one-shot review and 14–38% with event-driven review) and found no error of its own.

Table S7. Evaluation stages of the article, their design, registration and evidence levels (summarized in Section 5.4 of the article). Levels: A, Analyses prespecified in repository-recorded plans for the designated evaluation runs, following exploratory development; timing qualifications and deviations are reported in Table S10.; B, exploratory or descriptive; C, amendmentinformed; D, formal or deterministic verification; E, performance on one host (no device or multi-host measurement). Repository identifiers, configurations and seeds: Table S8.

Stage	Question	Design	Registration	Level
Main evaluation	Which pipelines admit forged or lose legitimate data?	E1–E15, eight pipelines, seeds 1–20 (1,500 + 620 runs)	P1–P3 for B5 and B5s; P4 (B4) describes stored runs; B1–B4 comparisons not registered	A (P1–P3), B
Mixed workload	All classes at once	16th workload, seeds 1–20 (60 runs)	exploratory	B
Re-execution with time-dependent outcomes	Are stored results reproducible; what do final counts conceal?	main evaluation and mixed workload re-run with the tracer (1,320 runs)	replication gate of the second registration	B
Review-schedule evaluation	Does the availability difference survive faster or matched review?	E5, E8, E11b, mixed; PBTI and 21 B5/PBTI-R schedule arms; seeds 21–40 (1,760 runs)	H1a–H1d; gates G1, G2	A
Selective-hold grid	Do targeted holds reduce retained forgeries?	equivocation × suppression grid, delayed successors, suppression runs; 7 arms; seeds 21–40 (2,940 runs)	H3a–H3c; gates G3, G4	A
Selective-hold factorial cells	Same, where feed-level review retained fewer forgeries	eight factorial cells, 7 arms, seeds 21–40 (1,120 runs)	amendment; gates G3F, G4F	C
Selective-hold controls	Side effects in every workload	16 workloads, seeds 1–20 (400 runs)	descriptive	B
Sensitivity analyses	Dependence on stressor rates	11-factor sweep at 0–20%, 2_{IV}^{7-3} factorial; seeds 1–20 (2,820 + 960 runs)	registered as descriptive	B
Public-telemetry replay	Real sensor dynamics	SKAB at 1, 5, 10×, seeds 1–20 (207 runs)	exploratory	B
Corrected-comparator reruns	Do the results depend on the known B5 defects?	every B5 and B5s specification of all stages above, original seeds (5,229 runs)	registered outcomes, families, margins and gates applied unchanged; correction decided after the results (DV8)	that of the rerun stage
Attacker-model robustness	How much do the results depend on the attack generator?	mixed workload with a consistent impersonator; PBTI, PBTI-R, B5 (one-shot and event-driven); seeds 21–40 (148 runs)	not registered (DV9)	B
Control traces	Which failure modes are common to all rules?	six deterministic traces, four pipelines	not registered	D
Lifecycle analysis	Which evidence justifies release under a revoked key?	68 deterministic timelines (11 cases, 3 policies, 2 pipelines)	not registered	D
Formal properties	Locality and signature bindings	proof, symbolic model, bounded enumeration	—	D
Performance	Processing, size and latency costs	in-process, batching, MQTT/TLS on one host	descriptive	E

Table S8. Evaluation stages of the article and their repository identifiers, configurations, seeds and run counts. All paths are relative to the repository root.

Stage (article)	Raw results	Configuration or command; registration	Seeds; runs
Main evaluation, B1–B4 and PBTI	`results/raw/scenarios*.jsonl`, `ablations*.jsonl`, `revocation*.jsonl`, `audit.jsonl`	`experiments/run_campaign.py` (`make campaign`); not registered	1–20; 1,500 scenario runs
Main evaluation, B5, B5s, PBTI-R	`results/raw/v3_scenarios_b5*.jsonl`, `v3_scenarios_pbtir.jsonl`	`make campaign-v3`; first registration `research/V3/PREREGISTRATION.md`	1–20; 620
Mixed workload	`results/raw/v3_mixed_b5*.jsonl`	as above; exploratory	1–20; 60
Re-execution with time-dependent outcomes	`results/v4/raw/A0_core.jsonl`; `results/v4/summary/A0_comparison.json`	`experiments/v4_configs/A0_core.json`; replication gate of the second registration	1–20; 1,320
Review-schedule evaluation	`results/v4/raw/A1_review.jsonl`	`experiments/v4_configs/A1_review.json`; second registration `research/V4/PREREGISTRATION_V4.md`	21–40; 1,760
Selective-hold grid	`results/v4/raw/A3_selective.jsonl`	`experiments/v4_configs/A3_selective.json`; second registration	21–40; 2,940
Selective-hold factorial cells	`results/v4/raw/A3_factorial.jsonl`	`experiments/v4_configs/amendment1/A3_factorial.json`; amendment `research/V4/PREREGISTRATION_V4_AMENDMENT_1.md`	21–40; 1,120
Selective-hold controls	`results/v4/raw/A3_scenarios.jsonl`	`experiments/v4_configs/A3_scenarios.json`; descriptive	1–20; 400
Sensitivity analyses	`results/raw/v3_sweep.jsonl`, `v3_factorial.jsonl`	`make campaign-v3`; registered as descriptive	1–20; 2,820, 960
Public-telemetry replay	`results/raw/v3_replay_skab.jsonl`	`make campaign-v3` with `SKAB_DIR`; exploratory	1–20; 207
Lifecycle analysis	`results/v4/summary/A5_lifecycle_matrix.json`	`scripts/a5_lifecycle_matrix.py`; not registered	–; 68 rows
Bounded checks	`results/v4/formal/`, `results/formal/`	`formal/v4_locality_lifecycle_check.py`, `formal/rule_l_bounded_check.py`, `formal/*.pv`	–
Performance	`results/raw/throughput*.jsonl`, `mqtt_e2e.jsonl`, `v3_batching.jsonl`, `v3_mqtt_e2e*.jsonl`	`make perf`, `make perf-v3`, `make e2e`, `make e2e-v3`	repetitions in Tables S35–S40
Second-environment re-execution	`results/v7/reexec_cloud/`	`tools/reexec/reproduce_subset.sh` (Table S11)	subsets; Table S11
Corrected-comparator reruns	`results/v9/raw/*_B5c.jsonl`, `*_B5Sc.jsonl`	`experiments/v9/run_all_b5c.sh`; frozen configurations; analysis `tools/v9/build_corrected_results.sh`	as the stages; 5,229
Attacker-model robustness and controls	`results/v9/raw/attacker_consistent_impersonator.jsonl`, `attacker_controls.json`	`experiments/v9/run_attacker_consistent.py`, `attacker_controls.py`	21–40; 148
Reference-model check of the corrected B5	`results/v9/b5c_refmodel/`	`tools/v9/b5c_refmodel_campaign.sh`	generator seeds; 14,280 comparisons

Table S9. Principal findings of the article and their evidence levels (levels as in Table S7).

Finding	Level	Evidence
Final forged admissions are identical for per-identity and feed-level containment in all 15 scenarios and in the four confirmatory workloads under every review schedule (by construction the feed-level comparator can only be lower than PBTI-R with the same reviewer)	A	article Table 4; Table S19
Final losses are identical in 14 of 15 scenarios; the exception is recovery of a beyond-horizon backlog by the reviewer, which PBTI-R matches	A	Table S19; article Section 6.2
With one-shot 60/10 s review, feed-level containment withholds genuine data longer (E5: +6.4 s per observation) but has less exposure in the mixed workload	A	article Table 5
With event-driven review, availability is equivalent in E5 and E8 and better for B5 in E11b and the mixed workload through recovery (gate G1 not met)	A	article Table 5 and Figure 2
With matched review, final outcomes are identical or equivalent in every schedule and workload (gate G2); withholding differs with $R \geq 5$ s or 10 s cadence, and feed holds reduce exposure in the mixed workload unless review is event-driven	A	article Section 6.3; Table S24
When equivocation coincides with suppressed messages, feed-level holds retain fewer forgeries than Rule L	B, C	article Section 6.4; Tables S31 and S26
Selective holds remove those forgeries at a cost in genuine data; their registered grid gates failed	A (negative), C	article Section 6.4; Figure S3; Table S27
A time-stamp condition for release under revoked keys admits backdated forgeries; an exact-content witness policy prevents them if the witness is trusted	D	article Section 6.6; Table S29
k admitted statements change the class and authority of at most $2k$ identities when no lifecycle transition intervenes (proof sketch; bounded checks)	D	article Proposition 1; Table S5
Most of W in E5 and in the factorial cells is end-of-run loss (observations still missing at the horizon); the extra withholding of the feed-level comparator is delay	B (exploratory)	article Section 6.1; Table S17
With a consistent impersonator every pipeline retains far more forgeries than with the registered one; one-shot feed holds still expose less and withhold more, and event-driven B5 and PBTI-R stay close	B (exploratory)	article Section 6.5; Tables S14–S16
Correcting the comparator's known defects changed no registered class, claim gate or prediction; it lowered B5's withholding and losses and raised its exposure slightly where its reviewer acts quickly or factorial stressors combine	that of the rerun stage (DV8)	article Section 5.6; Tables S12 and S13
Throughput 0.19× and message size 2.91× of a Zero-Trust baseline; networked saturation between 1000 and 1250 messages/s on one host	E	article Section 6.8; Tables S35 and S39

Comparator correction. Because the confirmatory comparisons set PBTI against B5, the defects found by the audit were corrected rather than only described. The historical implementation (`src/pbti/b5.py`), which produced every frozen result and is listed in the freeze files, is unchanged; the corrected implementation is a separate module (`src/pbti/b5_corrected.py`) with four changes. (C1) The feed-level response is triggered only by a receipt that adds new fork evidence: a second admitted version of its identity, an admitted successor link naming another digest for it, or a predecessor link naming none of the admitted versions of its predecessor; a successor that names one of the versions of an already contested predecessor no longer holds released observations again or restarts a hold after a resume. (C2) Every held committed observation has an open review case: a reviewed release of another reason whose re-run admission lands in a held feed continues as that observation's hold case (outcome `DEFERRED_TO_FEED_HOLD`) instead of being closed without a case; a pending case of another reason for the same wire becomes the hold case; every hold is recorded as a slot-state change, which the event-driven review follows; and a hold case opened while a review decision is applied is decided from the next periodic pass on. (C3) Holds are counted once per transition, and distinct held and recovered observations are counted separately. (C4) The release of a tainted held observation is labelled as such. Identity order within a pass, the absence of review cases for contested identities with resume checked after each hold-case decision, and the event-driven candidate rule are kept; the six points that the specification had left open are now stated there (`docs/BASELINES.md`). 12 regression tests reproduce each defect on the historical class and check the corrected behaviour, the kept choices and unchanged behaviour without fork evidence, and in 14,280 comparisons over the stream families of the reference model the corrected implementation matched the reference model with the corrected choices on the final authoritative set and every authority interval (14,280) and on the hold, fork and resume counters (14,280); it

differed from the historical implementation in 5,111 of these stress streams. Every B5 and B5s specification of the main evaluation, the mixed workload, the sweep, the factorial, the replay, the re-execution with the tracer, the review-schedule evaluation, the selective-hold grid and the factorial cells was rerun with the corrected implementation on its original seeds (5,229 of 5,229 runs); the frozen rows are kept, and the corrected rows are stored separately with the implementation identifier and the source digest. Historical runs without any fork evidence were reproduced in every outcome field (2,468 of 2,468). The unchanged analysis scripts were then run on a tree in which only the B5 and B5s rows are replaced; outcome definitions, comparison families, margins and gates are those of the registrations. Table S12 gives the run-level corrected-minus-original differences and Table S13 the comparisons whose class changed; Data S1 contains both versions of every per-cell table and comparison. The performance runs of B5 were not repeated: the in-process batching and worker study, whose disposition sequences equal those of PBTI, and the end-to-end MQTT runs, in which every receipt was admitted, contain no fork evidence, so the correction cannot change their dispositions, but their timings were not re-measured with the corrected module. The lifecycle matrix (Table S29) was recomputed with the corrected B5 and is unchanged (0 differing rows).

Table S10. Registration chronology, amendments and deviations. All plans are self-recorded in the repository with SHA-256 hashes and times written by the executing process; they were not deposited with an external registry, so a hash attests that a file is unchanged, not when it was written.

Record	Recorded (self-reported)	Content and effect
First registration (`research/V3/PREREGISTRATION.md`, SHA-256 `91f6b544...`)	3 Nov 2026, 22:51 UTC−6	outcomes O1, O2; predictions P1–P4; decision rules for comparative wording; written before any feed-level, sweep, release-workflow, batching or new end-to-end result existed
Amendments 1–3 to the first registration	3 Nov 2026, 23:26 UTC−6, before any campaign result was read	exploratory PBTI-R arm; full audits on seeds 1–5; 120 s production phase in the sensitivity analyses; no test or prediction changed
Amendment 4 to the first registration	4 Nov 2026, 01:06 UTC−6	B5 implementation fix after an audit failure in 2 of 20 mixed-workload runs, and re-run of every B5 and B5s arm; no test or prediction changed
Second registration (`research/V4/PREREGISTRATION_V4.md`, SHA-256 `9d3247ff...`, with the source and configuration hashes)	5 Nov 2026, 05:39 UTC, before any run on seeds 21–40	hypotheses H1a–H1d, H3a–H3c; outcomes; families; margins; classification rules; gates G1–G4; replication gate
Amendment to the second registration (SHA-256 `ac7945bb...`)	5 Nov 2026, 05:42 UTC, before any confirmatory selective-hold run	eight factorial cells with equivocation high and gates G3F, G4F; motivated by a one-seed pilot (seed 101) with retained forgeries at one grid point; data dependence disclosed
Deviation DV1	during the re-execution	host restart after 318 of 1,320 runs; resumed with the frozen runner; no effect expected (deterministic runs)
Deviation DV2	during the campaigns	campaigns run in a changed order; frozen source and configurations verified unchanged
Deviation DV3	after the results	descriptive summaries of the selective-hold grid added; no test, class, family, margin or gate changed
Deviation DV4	before the amendment	the selective-hold pilot was inspected beyond its error status, which led to the amendment
Deviation DV5	analysis	the re-execution comparison also excludes audit-outcome fields (reported separately) and two runner labels
Deviation DV6 (found by the post-hoc audit)	analysis	the review-schedule Holm families contain 16 tests the hypotheses do not name (H1a on O2; H1b on O1, O2 and NLX), so the families of O1 and NLX have 84 instead of 80 tests and that of O2 48 instead of 40; conservative; with the families of the text no class or significance changes
Deviation DV7 (found by the post-hoc audit)	analysis	the bootstrap generator is seeded with 20261104 and the CRC32 checksum of each comparison's key, a derivation the plan does not state; it changes intervals only by Monte-Carlo error, and one resampling-dependent class results (Table S27)
Timing (found by the post-hoc audit)	3–4 Nov 2026	the first registration was written after the B1–B4 comparisons with PBTI had been run (B4 scenario runs started at 17:57 UTC−6), so P4 describes stored runs and is reported as descriptive; B5 and its reviewer were debugged in smoke runs on seed 1, which the main evaluation also uses; amendments 1–3 were recorded at 23:26, eight minutes after the campaigns they govern were launched and before any result was read; the re-run with the corrected B5 started at 01:00, six minutes before amendment 4 was recorded
Timing (found by the post-hoc audit)	5 Nov 2026	the second registration records single-seed development values that already showed event-driven B5 close to PBTI in E5; the factorial cells of the amendment ran before the registered grid (DV2); the amendment allows presenting PBTI-SH as useful if either G4 or G4F holds
Deviation (found by the post-hoc audit)	analysis	the first registration says that pairs identical in every seed are not tested; the analysis enters them into the Holm correction with $p = 1$ (conservative)
Deviation DV8	6 Nov 2026, after all results above	B5 corrected in a separate module (`src/pbti/b5_corrected.py`; historical module unchanged) after the audit had found its defects; every B5 and B5s specification rerun on its original seeds and analysed with the unchanged scripts; the article's principal results use the corrected comparator, the historical results are kept
Deviation DV9	6 Nov 2026, after all results above	exploratory attacker-model campaign (consistent impersonator, seeds 21–40) and control traces added; not registered, classes descriptive
Note (found by the post-hoc audit)	analysis	the analysis script that classifies the comparisons was not among the hash-frozen files; a re-implementation written from the plan text reproduced 1180 of the 1182 classes, the other two (one grid point) depend on bootstrap resampling, and with the seeding of DV7 all 1182 were reproduced exactly

Table S11. Second-environment re-execution of selected seeds (all fields compared except wall-clock timing: fields containing `_us`, `wall_s`, `audit_static_s`, `audit_replay_s`).

Stage	Seeds	Runs	Identical	Different	Errors	Field comparisons
Main evaluation and mixed workload (re-executed)	1, 2	132	132	0	0	12,367
Review-schedule evaluation	21, 22	176	176	0	0	19,550
Selective-hold grid	21, 22	294	294	0	0	29,687
Selective-hold controls	1, 2	40	40	0	0	3,995
Selective-hold factorial cells	21, 22	112	112	0	0	12,282
Lifecycle analysis (all rows)	–	68	68	0	–	–
Bounded checks (all parts)	–	–	yes	–	–	–
Test suite	–	–	187 passed, 1 skipped, 0 failed			

Table S12. Corrected minus historical B5 and B5s, run by run on the same specifications and seeds. Changed: runs in which O1, O2, NLX, *W* or AV5 differs; the other columns give the range of the differences over all runs of the stage (NLX, *W* and AV5 only where the tracer ran). Counters (holds, distinct held and recovered observations) are not outcomes and are not counted here.

Stage	Pipeline	Runs	Changed	ΔO1	ΔO2	ΔNLX	Δ*W* (s)	ΔAV5
Main evaluation (15 scenarios, seeds 1–20)	B5	300	0	0	0	–	–	–
Main evaluation (15 scenarios, seeds 1–20)	B5s	300	0	0	0	–	–	–
Mixed workload (seeds 1–20)	B5	20	0	0	0	–	–	–
Mixed workload (seeds 1–20)	B5s	20	0	0	0	–	–	–
Stressor sweep	B5	940	0	0	0	–	–	–
Sensitivity factorial	B5	320	31	0	−4 to 0	–	–	–
SKAB replay	B5	69	0	0	0	–	–	–
Re-execution with the tracer	B5	320	0	0	0	0	0	0
Re-execution with the tracer	B5s	320	0	0	0	0	0	0
Review-schedule evaluation (seeds 21–40)	B5	880	202	0	−1 to 0	0.0 to 29.2	−0.057 to 0.000	0.0000 to 0.0032
Selective-hold grid	B5	840	171	0 to 1	−1 to 0	0.0 to 192.0	−0.101 to 0.000	0.0000 to 0.0003
Selective-hold grid	B5s	420	0	0	0	0	0	0
Selective-hold factorial cells	B5	320	184	0 to 1	−5 to 0	0.0 to 154.8	−0.289 to 0.000	0.0000 to 0.0014
Selective-hold factorial cells	B5s	160	0	0	0	0	0	0

Table S13. Registered comparisons whose class differs between the historical and the corrected comparator (0 of 1,182 comparisons of the review-schedule evaluation, the selective-hold grid and the factorial cells; 90 estimates changed at all). Mean paired difference (arm − reference) with 95% bootstrap interval.

Stage	Outcome	Cell	Comparison	Original: mean (95% CI), class	Corrected: mean (95% CI), class
No comparison changed its class.					

Table S14. Mixed workload with the registered and the consistent impersonator (means per run over seeds 21–40; about 7200 legitimate observations per run). O1: forged observations authoritative at the end; O2: legitimate observations missing at the end; NLX: non-legitimate exposure; W: mean withheld seconds per legitimate observation; AV5: share first available within 5 s. Unlabelled B5 rows: corrected implementation.

Pipeline	Attacker	O1	O2	NLX (record-s)	W (s)	AV5
PBTI	registered	4.5	330.9	3,417	22.81	0.847
PBTI	consistent	88.7	419.6	23,948	25.58	0.835
PBTI-$R_{60/10}$	registered	4.5	277.1	3,417	21.17	0.847
PBTI-$R_{60/10}$	consistent	88.7	364.9	23,948	23.90	0.835
PBTI-R_{ev}	registered	4.5	271.4	3,417	20.42	0.847
PBTI-R_{ev}	consistent	88.7	360.3	23,948	23.19	0.835
$B5_{60/10}$	registered	4.5	278.6	1,658	28.69	0.733
$B5_{60/10}$	consistent	88.3	366.0	18,035	30.35	0.734
$B5_{ev}$	registered	4.5	272.2	3,374	20.46	0.847
$B5_{ev}$	consistent	88.5	361.1	23,925	23.23	0.835
$B5_{60/10}$ (historical B5)	registered	4.5	278.6	1,658	28.69	0.733
$B5_{60/10}$ (historical B5)	consistent	88.3	366.0	18,035	30.35	0.734
$B5_{ev}$ (historical B5)	registered	4.5	272.3	3,374	20.47	0.847
$B5_{ev}$ (historical B5)	consistent	88.5	361.1	23,925	23.23	0.835

Attacker-model robustness campaign. The consistent impersonator of the earlier five-seed check (*Exploratory analyses* below; `experiments/v9/consistent_impersonator.py`, the same logic moved into an importable module) was run on all 20 seeds 21–40 of the mixed workload of the review-schedule evaluation, for PBTI, PBTI-R and B5 with one-shot ($R = 60$ s, $C = 10$ s) and event-driven review, B5 in its corrected and historical implementations, with the configurations of that evaluation unchanged (`experiments/v9/run_attacker_consistent.py`). Registered-attacker values are the frozen rows and, for the corrected B5, its rerun; re-running the registered attacker for seeds 21 and 22 reproduced those rows in 8 of 8 cases. The campaign is exploratory: it was designed after the registered results were known, and its classes use the registered margins descriptively, without family-wise correction (Tables S14 and S15). Table S16 lists the six deterministic control traces (`experiments/v9/attacker_controls.py`).

Table S15. Paired comparisons in the mixed workload under both attackers (difference = first minus second pipeline over seeds 21–40; B5: corrected implementation). Registered attacker: the registered comparison of Table S24 where one exists (its interval and class); all other entries: 95% percentile-bootstrap intervals and classes with the registered margins, used descriptively.

Comparison	Outcome	Registered attacker: mean (95% CI)	Consistent attacker: mean (95% CI)	Class (consistent)
$B5_{60/10}$ − PBTI	O1	0.0 [0.0, 0.0]	−0.3 [−0.6, −0.1]	equivalent
$B5_{60/10}$ − PBTI	NLX	−1,759.1 [−1,884.5, −1,637.2]	−5,912.4 [−6,993.3, −4,742.3]	better beyond margin
$B5_{60/10}$ − PBTI	W	+5.880 [5.729, 6.045]	+4.774 [4.667, 4.895]	worse beyond margin
$B5_{60/10}$ − PBTI	AV5	−0.114 [−0.116, −0.113]	−0.101 [−0.102, −0.100]	worse beyond margin
$B5_{ev}$ − PBTI	O1	0.0 [0.0, 0.0]	−0.1 [−0.3, 0.0]	equivalent
$B5_{ev}$ − PBTI	NLX	−42.7 [−58.9, −28.1]	−22.4 [−47.0, −0.7]	equivalent
$B5_{ev}$ − PBTI	W	−2.352 [−2.386, −2.314]	−2.348 [−2.388, −2.299]	better beyond margin
$B5_{ev}$ − PBTI	AV5	−0.000 [−0.001, −0.000]	−0.000 [−0.001, −0.000]	equivalent
$B5_{60/10}$ − PBTI-$R_{60/10}$	O1	0.0 [0.0, 0.0]	−0.3 [−0.6, −0.1]	equivalent
$B5_{60/10}$ − PBTI-$R_{60/10}$	NLX	−1,759.1 [−1,887.2, −1,637.4]	−5,912.4 [−6,988.1, −4,672.3]	better beyond margin
$B5_{60/10}$ − PBTI-$R_{60/10}$	W	+7.524 [7.377, 7.696]	+6.450 [6.361, 6.540]	worse beyond margin
$B5_{60/10}$ − PBTI-$R_{60/10}$	AV5	−0.114 [−0.116, −0.113]	−0.101 [−0.102, −0.100]	worse beyond margin
$B5_{ev}$ − PBTI-R_{ev}	O1	0.0 [0.0, 0.0]	−0.1 [−0.3, 0.0]	equivalent
$B5_{ev}$ − PBTI-R_{ev}	NLX	−42.7 [−59.3, −28.3]	−22.4 [−47.3, −0.8]	equivalent
$B5_{ev}$ − PBTI-R_{ev}	W	+0.043 [0.027, 0.061]	+0.041 [0.027, 0.057]	equivalent
$B5_{ev}$ − PBTI-R_{ev}	AV5	−0.000 [−0.001, −0.000]	−0.000 [−0.000, −0.000]	equivalent
PBTI-R_{ev} − PBTI	O1	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	identical
PBTI-R_{ev} − PBTI	NLX	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	identical
PBTI-R_{ev} − PBTI	W	−2.396 [−2.425, −2.364]	−2.389 [−2.425, −2.340]	better beyond margin
PBTI-R_{ev} − PBTI	AV5	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	identical

Table S16. Control traces on the real verifiers (one device, ten 1 Hz observations, simulated time to 200 s; the attacker holds the device key and relays through the honest gateway; PBTI-R and B5 with one-shot review, $R = 60$ s, $C = 10$ s). Forged and genuine counts are of distinct (identity, content) pairs that arrived; exposure and withheld time in record-seconds.

Trace	Pipeline	Forged authoritative at end	Forged exposure (record-s)	Genuine authoritative at end	Genuine withheld (record-s)
T1 forged version, contradicting successor admitted	PBTI	0/1	1.0	9/9	0.0
	PBTI-R	0/1	1.0	9/9	0.0
	B5 (historical)	0/1	1.0	9/9	255.9
	B5 (corrected)	0/1	1.0	9/9	255.9
T2 forged version, successors suppressed	PBTI	1/1	194.0	5/5	0.0
	PBTI-R	1/1	194.0	5/5	0.0
	B5 (historical)	1/1	194.0	5/5	0.0
	B5 (corrected)	1/1	194.0	5/5	0.0
T3 consistent takeover of the rest of the feed	PBTI	5/5	959.9	5/5	0.0
	PBTI-R	5/5	959.9	5/5	0.0
	B5 (historical)	5/5	959.9	5/5	0.0
	B5 (corrected)	5/5	959.9	5/5	0.0
T4 forgeries at identities the device does not use	PBTI	3/3	564.3	10/10	0.0
	PBTI-R	3/3	564.3	10/10	0.0
	B5 (historical)	3/3	564.3	10/10	0.0
	B5 (corrected)	3/3	564.3	10/10	0.0
T5 as T4, the device later reaches them	PBTI	0/3	93.4	11/15	629.9
	PBTI-R	0/3	93.4	11/15	629.9
	B5 (historical)	0/3	87.4	11/15	629.9
	B5 (corrected)	0/3	87.4	11/15	629.9
T6 forged epoch advance (drain 20 s)	PBTI	5/5	974.9	5/10	815.0
	PBTI-R	5/5	974.9	10/10	315.0
	B5 (historical)	5/5	974.9	10/10	315.0
	B5 (corrected)	5/5	974.9	10/10	315.0

Reference check (automated). Every entry of the article's reference list was checked automatically on 6 November 2026 against Crossref or DataCite, the RFC Editor (including errata), the IETF Datatracker or the issuing body; no retraction, correction notice or expression of concern was found. The statement each citation supports was checked against the cited work: from the full or standard text for 48 of the 65 entries, from the abstract for 9 and from catalogue records for 6 (these only identify a standard); the text of two works was not accessible, so AUPS is cited only for what its title states and Holm's procedure for its standard attribution. The C2PA specification is cited in version 2.5, which keeps the cited revocation rule.

Exploratory analyses added after the confirmatory analysis. Table S17 gives the split of W used in the article: PBTI, PBTI-R and PBTI-SH from the re-executed runs described next (their rows are unchanged), and B5 from the corrected reruns, where it is computed in each run from the run's own tracer with the same definitions and sums to the registered W in every run. (i) *Split of W (historical comparator).* 400 registered runs of seeds 21–40 (E5 and the mixed workload for PBTI, B5 with one-shot and with event-driven review and PBTI-R with event-driven review; the four factorial cells with equivocation and suppressed runs high for PBTI, PBTI-SH and $B5_{60/10}$) were re-executed with the oracle audit's capture, which reproduced the registered W in all of them, and W was split into the contribution of observations that are authoritative at the end of the run (delay) and of those that are not (end-of-run loss, not proof that later recovery is impossible). In E5, 99.7% of PBTI's W was end-of-run loss; $B5_{60/10}$ added 6.39 s of delay per observation and 0.0002 s of loss. In the mixed workload the delay was 15.90 s for $B5_{60/10}$ and 6.98 s for PBTI, the loss 12.79 and 15.83 s. In the factorial cells, PBTI-SH had 6.8 to 7.9 s per observation less delay than $B5_{60/10}$ and 0.7 to 2.7 s more end-of-run loss; 97–99% of PBTI-SH's W was end-of-run loss, against 71–85% for $B5_{60/10}$ (98% of PBTI-SH's withheld time in the cell examined by the oracle audit, counted until the end of the drain). (ii) *Consistent impersonator, five-seed check (historical comparator), superseded by the 20-seed campaign above.* Seeds 21–25 of the mixed workload were re-run for PBTI, $B5_{60/10}$, $B5_{ev}$ and PBTI-R_{ev} with the registered attacker, which reproduced the stored rows in all cases, and with an impersonator of the same capability that chains predecessor links by sequence number, resends one cached forgery per identity and drops transport duplicates in the impersonated range. Retained forgeries rose from 5.4 to 94.6 per run for PBTI (5.4 to 94.0 for $B5_{60/10}$), and exposure from 3,666 to 25,730 record-seconds for PBTI and from 1,981 to 20,056 for $B5_{60/10}$. Tools: `tools/v7/w_decomposition.py`, `tools/v7/consistent_impersonator_check.py`; outputs in `results/v7/v7_analysis/`.

Table S17. Split of the registered withheld time W into delay (observations authoritative at the end of the drain) and end-of-run loss (observations still missing then; not proof that later recovery is impossible), seconds per legitimate observation, means over seeds 21–40; denominator and horizon as for W. B5: corrected implementation; exploratory.

Workload	Pipeline	W (s)	Delay (s)	End-of-run loss (s)	Loss share
E5 stolen keys	PBTI	8.67	0.03	8.64	99.7%
	PBTI-R_{ev}	8.67	0.03	8.64	99.7%
	$B5_{60/10}$	15.06	6.42	8.64	57.4%
	$B5_{ev}$	8.67	0.04	8.64	99.6%
Mixed workload	PBTI	22.81	6.98	15.83	69.4%
	PBTI-R_{ev}	20.42	8.02	12.40	60.7%
	$B5_{60/10}$	28.69	15.90	12.79	44.6%
	$B5_{ev}$	20.46	8.03	12.43	60.8%
Cell run3	PBTI	20.46	0.25	20.21	98.8%
	PBTI-SH	21.60	0.38	21.22	98.2%
	$B5_{60/10}$	28.72	8.28	20.44	71.2%
	$B5_{ev}$	20.58	0.26	20.32	98.7%
Cell run4	PBTI	40.54	0.15	40.39	99.6%
	PBTI-SH	43.74	0.25	43.49	99.4%
	$B5_{60/10}$	47.86	7.03	40.83	85.3%
	$B5_{ev}$	40.85	0.16	40.69	99.6%
Cell run8	PBTI	22.69	0.24	22.45	98.9%
	PBTI-SH	23.76	0.37	23.39	98.4%
	$B5_{60/10}$	30.99	8.26	22.73	73.4%
	$B5_{ev}$	22.86	0.26	22.60	98.9%
Cell run15	PBTI	35.08	0.86	34.22	97.6%
	PBTI-SH	38.63	1.01	37.62	97.4%
	$B5_{60/10}$	43.32	8.23	35.09	81.0%
	$B5_{ev}$	35.62	0.89	34.73	97.5%

S4. Main Evaluation

Comparisons with B1–B4 (exploratory). Broker tampering (E2, E3) and timestamp-refreshed replays (E6) passed B1 and B2, whereas for the signing pipelines tampering became loss of the originals, which the broker had acknowledged; exact replays were suppressed by all pipelines. Duplicates, acknowledgement loss, reordering and partitions with reboot (E7, E9, E11/E12) were handled correctly by all pipelines; under clock faults (E10) deduplication on (device, timestamp) merged 68 legitimate observations per run in B1–B3, and key rotation during a partition (E13) cost B3 230. Of the 180 exploratory comparisons of PBTI with B1–B4, 118 were identical in every seed and 54 significant after Holm correction (Table S20).

In the re-execution with the tracer (Table S23), the tracer's final authoritative set matched the pipeline's in 1320 re-executed runs, and all 250 of 250 audited runs passed.

Table S18. Definitions of the time-dependent outcomes (repository `src/pbti/temporal.py`). $A(t)$: authoritative (event identity, content key) pairs; L: legitimate observations accepted by a gateway; interval $[0, T_{\text{end}}]$ = production plus drain (480 s in the scenarios, 300 s in the selective-hold stages), virtual time. One value per run.

Outcome	Definition	Unit
NLX	$\int \lvert A(t) \setminus \text{pairs}(L) \rvert \, dt$: exposure to non-legitimate authoritative pairs (a forgery retained to the end contributes until T_{end})	record-s
W	$\frac{1}{\lvert L \rvert} \sum_{o \in L} \left((T_{\text{end}} - t_{\text{acc}}(o)) - \lvert \{ t \geq t_{\text{acc}}(o) : o \text{ available} \} \rvert \right)$; includes end-of-run loss, so the contribution of an observation that is never made available grows with the drain	s per observation
AVd	share of L first available within $d \in \{1, 5, 30, 60\}$ s of production (also if later withdrawn)	fraction
KM p50/p90/p99	Kaplan–Meier quantiles of first availability; never-available observations censored at T_{end}	s
Withdrawn	observations available and later not available (once, or at the end)	count
Freshness	per asset at 1 s steps: no correct value available (missing) or age of the latest available correct value behind the latest produced (stale)	fraction, s
Containment latency	time from the first contradictory receipt (other content for the identity, or a successor whose prev contradicts it; any parseable receipt, also one that is rejected) or a revocation with t_{comp} to the end of a non-legitimate episode	s
Derived exposure	window-version time during which a derived window includes a non-legitimate input; time to the first version without it	s
Alerts	illustrative monitoring workload (not an application requirement): alert when temp_c > nominal +2.5 °C; true, false, missed (not raised within 5 s), retracted	count

Table S19. Main evaluation: non-legitimate authoritative observations (O1) and missing legitimate observations (O2), means over 20 paired seeds (7200 legitimate observations per run; E1 is zero for all pipelines). B5s: B5 without recovery. *: Holm-corrected Wilcoxon $p < 0.05$ versus PBTI in the registered family; †: held in the quarantine archive and releasable through review. Comparisons of B1–B4 with PBTI: Table S20. The 95% percentile-bootstrap CIs (10,000 resamples) of these and of the other per-scenario outcomes (reachable, quarantined or held legitimate observations, duplicate authoritative observations, corrupt and incomplete derived windows, current-value error, holds, releases and persisted bytes per observation) are provided as Data S1, file `v3_outcomes_full.csv`.

	Non-legitimate authoritative (mean/run)							**Legitimate missing** (mean/run)						
Scen.	B1	B2	B3	B4	B5	B5s	PBTI	B1	B2	B3	B4	B5	B5s	PBTI
E2	354.9	354.9	0	0	0	0	0	354.9	354.9	354.9	354.9	354.9	354.9	354.9
E3	427.6	225.7	0	0	0	0	0	427.9	427.9	427.7	427.7	427.7	427.7	427.7
E4	100	0	15.2	0	0	0	0	0	0	0	0	0	0	0
E5	298.9	298.9	298.9	289.8	150	150	150	198.8	198.8	198.8	288.9	222.6	929.7*	222.6
E6	212.1	212.1	0	0	0	0	0	0	0	0	0	0	0	0
E7	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E8	0	0	0	0	0	0	0	0	0	0	4	12	402*	12
E9	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E10	0	0	0	0	0	0	0	68	68	68	0	0	0	0
E11/12	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E11b	0	0	0	0	0	0	0	0	193.3	0	193.3	0*	193.3	193.3†
E13	0	0	0	0	0	0	0	0	0	230	0	0	0	0
E14	239.9	199.9	199.9	199.9	0	0	0	159.9	190.2	175.2	190.2	270.1	270.1	270.1
E15	100	100	15.8	15.8	15.8	15.8	15.8	90.0	90.0	89.5	89.5	89.5	89.5	89.5

Table S20. Two-sided Wilcoxon signed-rank tests of PBTI versus B1–B4 for non-legitimate authoritative observations, missing legitimate observations and current-value error in each scenario, paired by seed, Holm-corrected within each scenario and across all scenarios of a metric, with the matched-pairs rank-biserial correlation. 118 of the 180 comparisons were identical in every seed; with 20 pairs and no zero differences the smallest attainable exact two-sided p is 1.9×10^{-6} before correction. *Provided as Data S1, file* `v2_paired_tests.csv`.

Table S21. Feed-level holds, releases and latency in the main evaluation (means over 20 seeds; rows only where a pipeline forked, held, released or quarantined; last block: mixed workload). Forks: times a feed's fork point was set or lowered by new fork evidence; Held: hold transitions, one per observation entering the held set (an observation held again after later new fork evidence counts again; the historical counter also counted a hold twice when the held observation's own admission supplied the fork evidence, Section S3); Released: release events of held or quarantined observations by the automated reviewer; Lat.: time from production to authority of released observations (mean over runs of the per-run percentile); Quar.: legitimate observations missing at the end whose message had entered the quarantine archive or a feed hold, including observations in contested slots and in cases rejected by review, which cannot be released (Section S3); CV wrong: current-value error (does not penalize withheld data); B/obs.: persisted bytes per reachable observation. B5 here uses the one-shot 60/10 schedule (corrected implementation); B5s: B5 without recovery; PBTI-R: PBTI with the same reviewer (exploratory).

Scen.	Forks	Held	Released	Lat. p50 (s)	Lat. p95 (s)	Quar.	CV wrong %	B/obs.
E5 PBTI	0	0	0	–	–	0	3.09	1102
E5 B5	3	779.7	707.1	64.9	69.5	22.5	2.88	1272
E5 B5s	3	779.7	0	–	–	729.6	2.88	1100
E8 PBTI	0	0	0	–	–	0	0	1081
E8 B5	2	398	390	64.6	69.4	4	0	1171
E8 B5s	2	398	0	–	–	394	0	1080
E11b PBTI	0	0	0	–	–	193.3	0	1080
E11b B5	0	0	193.3	203.9	218.3	0	0	1132
E11b B5s	0	0	0	–	–	193.3	0	1080
E11b PBTI-R	0	0	193.3	203.9	218.3	0	0	1132
Mixed PBTI	0	0	0	–	–	60.1	0.76	1200
Mixed B5	6.0	978.4	886.8	65.7	190.3	37.2	0.23	1425
Mixed B5s	6.0	975.0	0	–	–	917.4	0.17	1200
Mixed PBTI-R	0	0	54.4	195.1	199.4	5.8	0.76	1219

Table S22. Registered comparisons of the main evaluation with a non-zero difference in at least one seed (two-sided Wilcoxon signed-rank on 20 paired seeds; Holm over the 30 tests of each outcome; positive difference: comparison pipeline larger). The other 57 of 60 tests were identical in every seed. Practical thresholds: 1 observation (O1), 72 observations (O2).

Outcome	Scen.	PBTI	Comp.	Diff. [95% CI]	r_{rb}	p_{Holm}	Practical
O2	E5 vs B5s	222.6	929.7	707.1 [699.3, 714.6]	1.00	0.0025	yes
O2	E8 vs B5s	12	402	390 [390, 390]	1.00	2.3×10^{-4}	yes
O2	E11b vs B5	193.3	0	-193.3 [-193.8, -192.9]	-1.00	0.002	yes

Table S23. Time-dependent outcomes of the main evaluation (re-execution with the tracer; means over 20 seeds). *W*: mean withheld seconds per legitimate observation; AV5: share first available within 5 s; KM p90: Kaplan–Meier 90th percentile of first availability (– if more than 10% are never available); Missing_{fresh}: share of 1 s samples without any correct available value. B5 uses the one-shot 60/10 schedule.

Scenario	Pipeline	O1	O2	NLX (rec-s)	*W* (s)	AV5	KM p90 (s)	Missing_{fresh}
E1	PBTI	0.0	0.0	0	0.03	1.000	0.0	0.000
	B4	0.0	0.0	0	0.03	1.000	0.0	0.000
	B5	0.0	0.0	0	0.03	1.000	0.0	0.000
	B5S	0.0	0.0	0	0.03	1.000	0.0	0.000
E2	PBTI	0.0	354.9	0	16.28	0.951	0.0	0.000
	B4	0.0	354.9	0	16.28	0.951	0.0	0.000
	B5	0.0	354.9	0	16.28	0.951	0.0	0.000
	B5S	0.0	354.9	0	16.28	0.951	0.0	0.000
E3	PBTI	0.0	427.7	0	19.55	0.941	0.1	0.000
	B4	0.0	427.7	0	19.55	0.941	0.1	0.000
	B5	0.0	427.7	0	19.55	0.941	0.1	0.000
	B5S	0.0	427.7	0	19.55	0.941	0.1	0.000
E4	PBTI	0.0	0.0	0	0.03	1.000	0.0	0.000
	B4	0.0	0.0	0	0.03	1.000	0.0	0.000
	B5	0.0	0.0	0	0.03	1.000	0.0	0.000
	B5S	0.0	0.0	0	0.03	1.000	0.0	0.000
E5	PBTI	150.0	222.6	38,266	8.64	0.972	0.0	0.000
	B4	289.8	288.9	84,353	11.32	0.960	0.0	0.000
	B5	150.0	222.6	38,242	15.02	0.871	62.9	0.000
	B5S	150.0	929.7	38,242	38.91	0.871	–	0.000
E6	PBTI	0.0	0.0	0	0.03	1.000	0.0	0.000
	B4	0.0	0.0	0	0.03	1.000	0.0	0.000
	B5	0.0	0.0	0	0.03	1.000	0.0	0.000
	B5S	0.0	0.0	0	0.03	1.000	0.0	0.000
E7	PBTI	0.0	0.0	0	0.07	1.000	0.0	0.000
	B4	0.0	0.0	0	0.07	1.000	0.0	0.000
	B5	0.0	0.0	0	0.07	1.000	0.0	0.000
	B5S	0.0	0.0	0	0.07	1.000	0.0	0.000
E8	PBTI	0.0	12.0	0	0.56	0.999	0.0	0.000
	B4	0.0	4.0	0	0.21	0.999	0.0	0.000
	B5	0.0	12.0	0	4.08	0.945	0.0	0.000
	B5S	0.0	402.0	0	15.74	0.945	0.0	0.000
E9	PBTI	0.0	0.0	0	0.47	1.000	2.0	0.001
	B4	0.0	0.0	0	0.47	1.000	2.0	0.001
	B5	0.0	0.0	0	0.47	1.000	2.0	0.001
	B5S	0.0	0.0	0	0.47	1.000	2.0	0.001
E10	PBTI	0.0	0.0	0	0.07	1.000	0.0	0.000
	B4	0.0	0.0	0	0.07	1.000	0.0	0.000
	B5	0.0	0.0	0	0.07	1.000	0.0	0.000
	B5S	0.0	0.0	0	0.07	1.000	0.0	0.000
E11/12	PBTI	0.0	0.0	0	6.76	0.889	17.5	0.000
	B4	0.0	0.0	0	6.76	0.889	17.5	0.000
	B5	0.0	0.0	0	6.76	0.889	17.5	0.000
	B5S	0.0	0.0	0	6.76	0.889	17.5	0.000
E11b	PBTI	0.0	193.3	0	17.54	0.867	40.9	0.000
	B4	0.0	193.3	0	17.54	0.867	40.9	0.000
	B5	0.0	0.0	0	12.17	0.867	40.9	0.000
	B5S	0.0	193.3	0	17.54	0.867	40.9	0.000
	PBTI-R	0.0	0.0	0	12.17	0.867	40.9	0.000
E13	PBTI	0.0	0.0	0	6.59	0.894	10.9	0.000
	B4	0.0	0.0	0	6.59	0.894	10.9	0.000
	B5	0.0	0.0	0	6.59	0.894	10.9	0.000
	B5S	0.0	0.0	0	6.59	0.894	10.9	0.000
E14	PBTI	0.0	270.1	8,042	11.41	0.974	0.0	0.000
	B4	199.9	190.2	64,028	8.30	0.974	0.0	0.000
	B5	0.0	270.1	8,042	11.41	0.974	0.0	0.000
	B5S	0.0	270.1	8,042	11.41	0.974	0.0	0.000
E15	PBTI	15.8	89.5	5,145	4.11	0.988	0.0	0.000
	B4	15.8	89.5	5,145	4.11	0.988	0.0	0.000
	B5	15.8	89.5	5,145	4.11	0.988	0.0	0.000
	B5S	15.8	89.5	5,145	4.11	0.988	0.0	0.000
Mixed	PBTI	5.2	335.0	3,462	22.95	0.847	56.6	0.000
	B4	147.7	327.6	48,052	22.37	0.845	59.4	0.000
	B5	5.1	282.1	1,829	28.84	0.733	69.5	0.000
	B5S	5.0	1,162.3	1,779	57.43	0.733	–	0.000
	PBTI-R	5.2	280.6	3,462	21.29	0.847	56.6	0.000

S5. Confirmatory Evaluation

Review-schedule evaluation. 1760 runs completed with 0 errors; the tracer matched in 1760 runs and 176 of 176 audited runs passed. Notation of the schedules as in the introduction to this document.

Table S24. Review-schedule evaluation (seeds 21–40): every registered comparison, $\bar{d}$ = mean paired difference (arm − reference) with 95% and 90% bootstrap CIs, Holm-adjusted Wilcoxon p (within each outcome family) and the mechanical classification of the registered plan. Outcomes: O1, O2 (observations per run), NLX (record-s), W (s per observation), AV5 (fraction). Classes: id, identical in every seed; eq, equivalent within the margin (90% CI inside ±margin); eq^s, equivalent and significant after Holm correction; **B**/**W**, better/worse beyond the margin (Holm-adjusted $p < 0.05$ and 95% CI beyond the margin); dw, significant, margin unresolved (Holm-adjusted $p < 0.05$, 95% CI crossing a margin boundary; historical label "different within the margin"); inc, inconclusive.

Hyp.	Scen.	Arm	Reference	Out.	$\bar{d}$ 95% CI	90% CI	p_{Holm}	Class
H1a	E5	$B5_{60/10}$	PBTI	AV5	−0.102	[−0.103, −0.101]	[−0.102, −0.101]	0.003	**W**
H1a	E5	$B5_{60/10}$	PBTI	NLX	−22.1	[−24.2, −20]	[−23.8, −20.3]	<0.001	eq^s
H1a	E5	$B5_{60/10}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1a	E5	$B5_{60/10}$	PBTI	O2	0	[0, 0]	[0, 0]	1.000	id
H1a	E5	$B5_{60/10}$	PBTI	W	6.39	[6.34, 6.45]	[6.34, 6.44]	<0.001	**W**
H1a	E8	$B5_{60/10}$	PBTI	AV5	−0.0547	[−0.0547, −0.0547]	[−0.0547, −0.0547]	<0.001	**W**
H1a	E8	$B5_{60/10}$	PBTI	NLX	0	[0, 0]	[0, 0]	1.000	id
H1a	E8	$B5_{60/10}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1a	E8	$B5_{60/10}$	PBTI	O2	0	[0, 0]	[0, 0]	1.000	id
H1a	E8	$B5_{60/10}$	PBTI	W	3.52	[3.52, 3.52]	[3.52, 3.52]	<0.001	**W**
H1a	E11b	$B5_{60/10}$	PBTI	AV5	0	[0, 0]	[0, 0]	1.000	id
H1a	E11b	$B5_{60/10}$	PBTI	NLX	0	[0, 0]	[0, 0]	1.000	id
H1a	E11b	$B5_{60/10}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1a	E11b	$B5_{60/10}$	PBTI	O2	−193	[−194, −193]	[−194, −193]	0.004	**B**
H1a	E11b	$B5_{60/10}$	PBTI	W	−5.37	[−5.38, −5.35]	[−5.38, −5.36]	0.001	**B**
H1a	Mixed	$B5_{60/10}$	PBTI	AV5	−0.114	[−0.116, −0.113]	[−0.115, −0.113]	0.003	**W**
H1a	Mixed	$B5_{60/10}$	PBTI	NLX	−1,759	[−1,885, −1,637]	[−1,866, −1,657]	<0.001	**B**
H1a	Mixed	$B5_{60/10}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1a	Mixed	$B5_{60/10}$	PBTI	O2	−52.3	[−55, −49.4]	[−54.5, −49.9]	0.004	eq^s
H1a	Mixed	$B5_{60/10}$	PBTI	W	5.88	[5.73, 6.04]	[5.75, 6.02]	<0.001	**W**
H1b	E5	$B5_{ev}$	PBTI	AV5	$-3.3 \cdot 10^{-4}$	$[-4.2 \cdot 10^{-4}, -2.4 \cdot 10^{-4}]$	$[-4.1 \cdot 10^{-4}, -2.6 \cdot 10^{-4}]$	0.005	eq^s
H1b	E5	$B5_{ev}$	PBTI	NLX	−1.9	[−2.49, −1.36]	[−2.39, −1.45]	0.007	eq^s
H1b	E5	$B5_{ev}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1b	E5	$B5_{ev}$	PBTI	O2	0	[0, 0]	[0, 0]	1.000	id
H1b	E5	$B5_{ev}$	PBTI	W	0.00861	[0.00821, 0.009]	[0.00829, 0.00894]	<0.001	eq^s
H1b	E8	$B5_{ev}$	PBTI	AV5	0	[0, 0]	[0, 0]	1.000	id
H1b	E8	$B5_{ev}$	PBTI	NLX	0	[0, 0]	[0, 0]	1.000	id
H1b	E8	$B5_{ev}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1b	E8	$B5_{ev}$	PBTI	O2	0	[0, 0]	[0, 0]	1.000	id
H1b	E8	$B5_{ev}$	PBTI	W	$5.5 \cdot 10^{-4}$	$[5.5 \cdot 10^{-4}, 5.6 \cdot 10^{-4}]$	$[5.5 \cdot 10^{-4}, 5.6 \cdot 10^{-4}]$	0.001	eq^s
H1b	E11b	$B5_{ev}$	PBTI	AV5	0	[0, 0]	[0, 0]	1.000	id
H1b	E11b	$B5_{ev}$	PBTI	NLX	0	[0, 0]	[0, 0]	1.000	id
H1b	E11b	$B5_{ev}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1b	E11b	$B5_{ev}$	PBTI	O2	−193	[−194, −193]	[−194, −193]	0.004	**B**
H1b	E11b	$B5_{ev}$	PBTI	W	−7.22	[−7.24, −7.21]	[−7.24, −7.21]	<0.001	**B**
H1b	Mixed	$B5_{ev}$	PBTI	AV5	$-4.2 \cdot 10^{-4}$	$[-5.5 \cdot 10^{-4}, -3.1 \cdot 10^{-4}]$	$[-5.3 \cdot 10^{-4}, -3.3 \cdot 10^{-4}]$	0.003	eq^s
H1b	Mixed	$B5_{ev}$	PBTI	NLX	−42.7	[−58.9, −28.1]	[−56.2, −30.3]	<0.001	eq^s
H1b	Mixed	$B5_{ev}$	PBTI	O1	0	[0, 0]	[0, 0]	1.000	id
H1b	Mixed	$B5_{ev}$	PBTI	O2	−58.8	[−59.6, −57.8]	[−59.5, −58]	0.004	eq^s
H1b	Mixed	$B5_{ev}$	PBTI	W	−2.35	[−2.39, −2.31]	[−2.38, −2.32]	<0.001	**B**
H1c	E5	$B5_{D0/1}$	PBTI-$R_{D0/1}$	AV5	$-5.7 \cdot 10^{-4}$	$[-6.9 \cdot 10^{-4}, -4.5 \cdot 10^{-4}]$	$[-6.7 \cdot 10^{-4}, -4.7 \cdot 10^{-4}]$	0.003	eq^s
H1c	E5	$B5_{D0/1}$	PBTI-$R_{D0/1}$	NLX	−2.67	[−3.55, −1.91]	[−3.41, −2.02]	<0.001	eq^s
H1c	E5	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D0/1}$	PBTI-$R_{D0/1}$	W	0.0154	[0.0141, 0.0166]	[0.0143, 0.0164]	<0.001	eq^s
H1c	E5	$B5_{D0/10}$	PBTI-$R_{D0/10}$	AV5	−0.0164	[−0.0174, −0.0154]	[−0.0172, −0.0155]	0.003	**W**
H1c	E5	$B5_{D0/10}$	PBTI-$R_{D0/10}$	NLX	−9.57	[−11.2, −8.05]	[−10.9, −8.29]	<0.001	eq^s
H1c	E5	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D0/10}$	PBTI-$R_{D0/10}$	W	0.168	[0.159, 0.177]	[0.161, 0.175]	<0.001	eq^s
H1c	E5	$B5_{D1/1}$	PBTI-$R_{D1/1}$	AV5	−0.0033	[−0.00353, −0.00306]	[−0.0035, −0.0031]	0.003	eq^s

Hyp.	Scen.	Arm	Reference	Out.	$\bar{d}$	95% CI	90% CI	p_{Holm}	Class
H1c	E5	$B5_{D1/1}$	PBTI-$R_{D1/1}$	NLX	−21.2	[−23.7, −18.7]	[−23.3, −19.1]	<0.001	eq^s
H1c	E5	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D1/1}$	PBTI-$R_{D1/1}$	W	0.141	[0.134, 0.149]	[0.135, 0.147]	<0.001	eq^s
H1c	E5	$B5_{D1/10}$	PBTI-$R_{D1/10}$	AV5	−0.0622	[−0.063, −0.0614]	[−0.0629, −0.0615]	0.003	**W**
H1c	E5	$B5_{D1/10}$	PBTI-$R_{D1/10}$	NLX	−22	[−24.1, −19.9]	[−23.8, −20.3]	<0.001	eq^s
H1c	E5	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D1/10}$	PBTI-$R_{D1/10}$	W	0.583	[0.573, 0.593]	[0.574, 0.591]	<0.001	eq^s
H1c	E5	$B5_{D5/1}$	PBTI-$R_{D5/1}$	AV5	−0.102	[−0.103, −0.101]	[−0.102, −0.101]	0.003	**W**
H1c	E5	$B5_{D5/1}$	PBTI-$R_{D5/1}$	NLX	−22.1	[−24.2, −20]	[−23.9, −20.4]	<0.001	eq^s
H1c	E5	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D5/1}$	PBTI-$R_{D5/1}$	W	0.537	[0.528, 0.545]	[0.529, 0.544]	<0.001	eq^s
H1c	E5	$B5_{D5/10}$	PBTI-$R_{D5/10}$	AV5	−0.102	[−0.103, −0.101]	[−0.102, −0.101]	0.003	**W**
H1c	E5	$B5_{D5/10}$	PBTI-$R_{D5/10}$	NLX	−22.1	[−24.3, −20]	[−23.9, −20.3]	<0.001	eq^s
H1c	E5	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D5/10}$	PBTI-$R_{D5/10}$	W	0.981	[0.97, 0.994]	[0.972, 0.992]	<0.001	**W**
H1c	E5	$B5_{D60/1}$	PBTI-$R_{D60/1}$	AV5	−0.102	[−0.103, −0.101]	[−0.102, −0.101]	0.003	**W**
H1c	E5	$B5_{D60/1}$	PBTI-$R_{D60/1}$	NLX	−22.1	[−24.3, −20]	[−23.9, −20.4]	<0.001	eq^s
H1c	E5	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D60/1}$	PBTI-$R_{D60/1}$	W	5.95	[5.9, 6]	[5.91, 5.99]	<0.001	**W**
H1c	E5	$B5_{D60/10}$	PBTI-$R_{D60/10}$	AV5	−0.102	[−0.103, −0.101]	[−0.102, −0.101]	0.003	**W**
H1c	E5	$B5_{D60/10}$	PBTI-$R_{D60/10}$	NLX	−22.1	[−24.2, −20]	[−23.9, −20.3]	<0.001	eq^s
H1c	E5	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{D60/10}$	PBTI-$R_{D60/10}$	W	6.39	[6.33, 6.44]	[6.34, 6.44]	<0.001	**W**
H1c	E5	$B5_{ev}$	PBTI-R_{ev}	AV5	$-3.3 \cdot 10^{-4}$	$[-4.2 \cdot 10^{-4}, -2.4 \cdot 10^{-4}]$	$[-4.1 \cdot 10^{-4}, -2.6 \cdot 10^{-4}]$	0.005	eq^s
H1c	E5	$B5_{ev}$	PBTI-R_{ev}	NLX	−1.9	[−2.45, −1.4]	[−2.36, −1.45]	0.007	eq^s
H1c	E5	$B5_{ev}$	PBTI-R_{ev}	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{ev}$	PBTI-R_{ev}	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{ev}$	PBTI-R_{ev}	W	0.00861	[0.00823, 0.009]	[0.00829, 0.00894]	<0.001	eq^s
H1c	E5	$B5_{60/10}$	PBTI-$R_{60/10}$	AV5	−0.102	[−0.103, −0.101]	[−0.102, −0.101]	0.003	**W**
H1c	E5	$B5_{60/10}$	PBTI-$R_{60/10}$	NLX	−22.1	[−24.2, −20]	[−23.8, −20.3]	<0.001	eq^s
H1c	E5	$B5_{60/10}$	PBTI-$R_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{60/10}$	PBTI-$R_{60/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E5	$B5_{60/10}$	PBTI-$R_{60/10}$	W	6.39	[6.34, 6.45]	[6.34, 6.44]	<0.001	**W**
H1c	E8	$B5_{D0/1}$	PBTI-$R_{D0/1}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/1}$	PBTI-$R_{D0/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/1}$	PBTI-$R_{D0/1}$	W	0.00113	[0.00105, 0.00121]	[0.00106, 0.00119]	<0.001	eq^s
H1c	E8	$B5_{D0/10}$	PBTI-$R_{D0/10}$	AV5	−0.00167	[−0.00167, −0.00167]	[−0.00167, −0.00167]	<0.001	eq^s
H1c	E8	$B5_{D0/10}$	PBTI-$R_{D0/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D0/10}$	PBTI-$R_{D0/10}$	W	0.0178	[0.0175, 0.0182]	[0.0176, 0.0181]	<0.001	eq^s
H1c	E8	$B5_{D1/1}$	PBTI-$R_{D1/1}$	AV5	$-2.6 \cdot 10^{-4}$	$[-2.8 \cdot 10^{-4}, -2.4 \cdot 10^{-4}]$	$[-2.8 \cdot 10^{-4}, -2.5 \cdot 10^{-4}]$	<0.001	eq^s
H1c	E8	$B5_{D1/1}$	PBTI-$R_{D1/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D1/1}$	PBTI-$R_{D1/1}$	W	0.0475	[0.0449, 0.0499]	[0.0453, 0.0495]	<0.001	eq^s
H1c	E8	$B5_{D1/10}$	PBTI-$R_{D1/10}$	AV5	−0.0325	[−0.0325, −0.0325]	[−0.0325, −0.0325]	<0.001	**W**
H1c	E8	$B5_{D1/10}$	PBTI-$R_{D1/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D1/10}$	PBTI-$R_{D1/10}$	W	0.322	[0.318, 0.326]	[0.319, 0.325]	<0.001	eq^s
H1c	E8	$B5_{D5/1}$	PBTI-$R_{D5/1}$	AV5	−0.0547	[−0.0547, −0.0547]	[−0.0547, −0.0547]	<0.001	**W**
H1c	E8	$B5_{D5/1}$	PBTI-$R_{D5/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O1	0	[0, 0]	[0, 0]	1.000	id

Hyp.	Scen.	Arm	Reference	Out.	$\bar{d}$	95% CI	90% CI	p_{Holm}	Class
H1c	E8	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D5/1}$	PBTI-$R_{D5/1}$	W	0.3	[0.296, 0.303]	[0.296, 0.303]	<0.001	eq^s
H1c	E8	$B5_{D5/10}$	PBTI-$R_{D5/10}$	AV5	−0.0547	[−0.0547, −0.0547]	[−0.0547, −0.0547]	<0.001	**W**
H1c	E8	$B5_{D5/10}$	PBTI-$R_{D5/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D5/10}$	PBTI-$R_{D5/10}$	W	0.544	[0.54, 0.548]	[0.541, 0.547]	<0.001	eq^s
H1c	E8	$B5_{D60/1}$	PBTI-$R_{D60/1}$	AV5	−0.0547	[−0.0547, −0.0547]	[−0.0547, −0.0547]	<0.001	**W**
H1c	E8	$B5_{D60/1}$	PBTI-$R_{D60/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D60/1}$	PBTI-$R_{D60/1}$	W	3.28	[3.27, 3.28]	[3.28, 3.28]	<0.001	**W**
H1c	E8	$B5_{D60/10}$	PBTI-$R_{D60/10}$	AV5	−0.0547	[−0.0547, −0.0547]	[−0.0547, −0.0547]	<0.001	**W**
H1c	E8	$B5_{D60/10}$	PBTI-$R_{D60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{D60/10}$	PBTI-$R_{D60/10}$	W	3.52	[3.52, 3.52]	[3.52, 3.52]	<0.001	**W**
H1c	E8	$B5_{ev}$	PBTI-R_{ev}	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{ev}$	PBTI-R_{ev}	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{ev}$	PBTI-R_{ev}	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{ev}$	PBTI-R_{ev}	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{ev}$	PBTI-R_{ev}	W	$5.5 \cdot 10^{-4}$	$[5.5 \cdot 10^{-4}, 5.6 \cdot 10^{-4}]$	$[5.5 \cdot 10^{-4}, 5.6 \cdot 10^{-4}]$	0.001	eq^s
H1c	E8	$B5_{60/10}$	PBTI-$R_{60/10}$	AV5	−0.0547	[−0.0547, −0.0547]	[−0.0547, −0.0547]	<0.001	**W**
H1c	E8	$B5_{60/10}$	PBTI-$R_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{60/10}$	PBTI-$R_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{60/10}$	PBTI-$R_{60/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E8	$B5_{60/10}$	PBTI-$R_{60/10}$	W	3.52	[3.52, 3.52]	[3.52, 3.52]	<0.001	**W**
H1c	E11b	$B5_{D0/1}$	PBTI-$R_{D0/1}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/1}$	PBTI-$R_{D0/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/1}$	PBTI-$R_{D0/1}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/10}$	PBTI-$R_{D0/10}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/10}$	PBTI-$R_{D0/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D0/10}$	PBTI-$R_{D0/10}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/1}$	PBTI-$R_{D1/1}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/1}$	PBTI-$R_{D1/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/1}$	PBTI-$R_{D1/1}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/10}$	PBTI-$R_{D1/10}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/10}$	PBTI-$R_{D1/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D1/10}$	PBTI-$R_{D1/10}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/1}$	PBTI-$R_{D5/1}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/1}$	PBTI-$R_{D5/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/1}$	PBTI-$R_{D5/1}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/10}$	PBTI-$R_{D5/10}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/10}$	PBTI-$R_{D5/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D5/10}$	PBTI-$R_{D5/10}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/1}$	PBTI-$R_{D60/1}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/1}$	PBTI-$R_{D60/1}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/1}$	PBTI-$R_{D60/1}$	W	0	[0, 0]	[0, 0]	1.000	id

Hyp.	Scen.	Arm	Reference	Out.	$\bar{d}$	95% CI	90% CI	p_{Holm}	Class
H1c	E11b	$B5_{D60/10}$	PBTI-$R_{D60/10}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/10}$	PBTI-$R_{D60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{D60/10}$	PBTI-$R_{D60/10}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{ev}$	PBTI-R_{ev}	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{ev}$	PBTI-R_{ev}	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{ev}$	PBTI-R_{ev}	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{ev}$	PBTI-R_{ev}	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{ev}$	PBTI-R_{ev}	W	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{60/10}$	PBTI-$R_{60/10}$	AV5	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{60/10}$	PBTI-$R_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{60/10}$	PBTI-$R_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{60/10}$	PBTI-$R_{60/10}$	O2	0	[0, 0]	[0, 0]	1.000	id
H1c	E11b	$B5_{60/10}$	PBTI-$R_{60/10}$	W	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D0/1}$	PBTI-$R_{D0/1}$	AV5	−0.00113	$[-0.00161, -7.3 \cdot 10^{-4}]$	$[-0.00152, -7.9 \cdot 10^{-4}]$	0.003	eq^s
H1c	Mixed	$B5_{D0/1}$	PBTI-$R_{D0/1}$	NLX	−72.5	[−89.6, −56.1]	[−86.5, −58.7]	<0.001	dw
H1c	Mixed	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D0/1}$	PBTI-$R_{D0/1}$	O2	0.8	[0.4, 1.25]	[0.45, 1.15]	0.153	eq
H1c	Mixed	$B5_{D0/1}$	PBTI-$R_{D0/1}$	W	0.0639	[0.0449, 0.0845]	[0.0477, 0.081]	<0.001	eq^s
H1c	Mixed	$B5_{D0/10}$	PBTI-$R_{D0/10}$	AV5	−0.0298	[−0.0324, −0.0272]	[−0.032, −0.0276]	<0.001	**W**
H1c	Mixed	$B5_{D0/10}$	PBTI-$R_{D0/10}$	NLX	−318	[−372, −266]	[−362, −274]	<0.001	**B**
H1c	Mixed	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D0/10}$	PBTI-$R_{D0/10}$	O2	1.05	[0.699, 1.45]	[0.75, 1.35]	0.027	eq^s
H1c	Mixed	$B5_{D0/10}$	PBTI-$R_{D0/10}$	W	0.372	[0.334, 0.41]	[0.341, 0.404]	<0.001	eq^s
H1c	Mixed	$B5_{D1/1}$	PBTI-$R_{D1/1}$	AV5	−0.00237	[−0.00278, −0.00201]	[−0.00271, −0.00206]	0.003	eq^s
H1c	Mixed	$B5_{D1/1}$	PBTI-$R_{D1/1}$	NLX	−152	[−175, −131]	[−171, −134]	<0.001	**B**
H1c	Mixed	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D1/1}$	PBTI-$R_{D1/1}$	O2	0.95	[0.6, 1.35]	[0.65, 1.25]	0.039	eq^s
H1c	Mixed	$B5_{D1/1}$	PBTI-$R_{D1/1}$	W	0.151	[0.13, 0.171]	[0.133, 0.168]	<0.001	eq^s
H1c	Mixed	$B5_{D1/10}$	PBTI-$R_{D1/10}$	AV5	−0.0651	[−0.0668, −0.0635]	[−0.0665, −0.0638]	0.003	**W**
H1c	Mixed	$B5_{D1/10}$	PBTI-$R_{D1/10}$	NLX	−406	[−456, −359]	[−449, −366]	<0.001	**B**
H1c	Mixed	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D1/10}$	PBTI-$R_{D1/10}$	O2	1.15	[0.75, 1.55]	[0.8, 1.5]	0.020	eq^s
H1c	Mixed	$B5_{D1/10}$	PBTI-$R_{D1/10}$	W	0.692	[0.665, 0.718]	[0.669, 0.714]	<0.001	**W**
H1c	Mixed	$B5_{D5/1}$	PBTI-$R_{D5/1}$	AV5	−0.114	[−0.116, −0.113]	[−0.115, −0.113]	0.003	**W**
H1c	Mixed	$B5_{D5/1}$	PBTI-$R_{D5/1}$	NLX	−399	[−454, −351]	[−445, −358]	<0.001	**B**
H1c	Mixed	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D5/1}$	PBTI-$R_{D5/1}$	O2	1.15	[0.75, 1.55]	[0.8, 1.5]	0.020	eq^s
H1c	Mixed	$B5_{D5/1}$	PBTI-$R_{D5/1}$	W	0.699	[0.675, 0.725]	[0.679, 0.721]	<0.001	**W**
H1c	Mixed	$B5_{D5/10}$	PBTI-$R_{D5/10}$	AV5	−0.114	[−0.116, −0.113]	[−0.115, −0.113]	0.003	**W**
H1c	Mixed	$B5_{D5/10}$	PBTI-$R_{D5/10}$	NLX	−610	[−697, −542]	[−681, −550]	<0.001	**B**
H1c	Mixed	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D5/10}$	PBTI-$R_{D5/10}$	O2	1.15	[0.75, 1.55]	[0.8, 1.5]	0.020	eq^s
H1c	Mixed	$B5_{D5/10}$	PBTI-$R_{D5/10}$	W	1.23	[1.19, 1.27]	[1.2, 1.26]	<0.001	**W**
H1c	Mixed	$B5_{D60/1}$	PBTI-$R_{D60/1}$	AV5	−0.114	[−0.116, −0.113]	[−0.115, −0.113]	0.003	**W**
H1c	Mixed	$B5_{D60/1}$	PBTI-$R_{D60/1}$	NLX	−1,721	[−1,840, −1,604]	[−1,822, −1,623]	<0.001	**B**
H1c	Mixed	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D60/1}$	PBTI-$R_{D60/1}$	O2	1.15	[0.75, 1.55]	[0.85, 1.5]	0.020	eq^s
H1c	Mixed	$B5_{D60/1}$	PBTI-$R_{D60/1}$	W	7.01	[6.87, 7.18]	[6.89, 7.15]	<0.001	**W**
H1c	Mixed	$B5_{D60/10}$	PBTI-$R_{D60/10}$	AV5	−0.114	[−0.116, −0.113]	[−0.115, −0.113]	0.003	**W**
H1c	Mixed	$B5_{D60/10}$	PBTI-$R_{D60/10}$	NLX	−1,759	[−1,885, −1,641]	[−1,864, −1,659]	<0.001	**B**
H1c	Mixed	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{D60/10}$	PBTI-$R_{D60/10}$	O2	1.15	[0.75, 1.55]	[0.8, 1.5]	0.020	eq^s
H1c	Mixed	$B5_{D60/10}$	PBTI-$R_{D60/10}$	W	7.52	[7.36, 7.69]	[7.39, 7.66]	<0.001	**W**
H1c	Mixed	$B5_{ev}$	PBTI-R_{ev}	AV5	$-4.2 \cdot 10^{-4}$	$[-5.5 \cdot 10^{-4}, -3.1 \cdot 10^{-4}]$	$[-5.3 \cdot 10^{-4}, -3.3 \cdot 10^{-4}]$	0.003	eq^s
H1c	Mixed	$B5_{ev}$	PBTI-R_{ev}	NLX	−42.7	[−59.3, −28.3]	[−56.3, −30.3]	<0.001	eq^s
H1c	Mixed	$B5_{ev}$	PBTI-R_{ev}	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{ev}$	PBTI-R_{ev}	O2	0.8	[0.4, 1.2]	[0.45, 1.15]	0.153	eq
H1c	Mixed	$B5_{ev}$	PBTI-R_{ev}	W	0.0433	[0.0269, 0.0612]	[0.0291, 0.0579]	<0.001	eq^s
H1c	Mixed	$B5_{60/10}$	PBTI-$R_{60/10}$	AV5	−0.114	[−0.116, −0.113]	[−0.115, −0.113]	0.003	**W**
H1c	Mixed	$B5_{60/10}$	PBTI-$R_{60/10}$	NLX	−1,759	[−1,887, −1,637]	[−1,866, −1,655]	<0.001	**B**

Hyp.	Scen.	Arm	Reference	Out.	$\bar{d}$	95% CI	90% CI	p_{Holm}	Class
H1c	Mixed	$B5_{60/10}$	PBTI-$R_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1c	Mixed	$B5_{60/10}$	PBTI-$R_{60/10}$	O2	1.5	[1, 2.05]	[1.05, 1.95]	0.022	eq^s
H1c	Mixed	$B5_{60/10}$	PBTI-$R_{60/10}$	W	7.52	[7.38, 7.7]	[7.4, 7.66]	<0.001	**W**
H1d	E5	$B5_{D0/1}$	$B5_{60/10}$	NLX	19.4	[17.8, 20.9]	[18.1, 20.7]	<0.001	eq^s
H1d	E5	$B5_{D0/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D0/10}$	$B5_{60/10}$	NLX	12.5	[11.5, 13.4]	[11.7, 13.3]	0.005	eq^s
H1d	E5	$B5_{D0/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D1/1}$	$B5_{60/10}$	NLX	0.85	[0, 2.5]	[0, 2.4]	1.000	eq
H1d	E5	$B5_{D1/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D1/10}$	$B5_{60/10}$	NLX	0.0996	[0, 0.299]	[0, 0.299]	1.000	eq
H1d	E5	$B5_{D1/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D5/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D5/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D5/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D5/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D60/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D60/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D60/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{D60/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E5	$B5_{ev}$	$B5_{60/10}$	NLX	20.2	[18.4, 22]	[18.7, 21.7]	<0.001	eq^s
H1d	E5	$B5_{ev}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D0/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D0/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D0/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D0/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D1/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D1/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D1/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D1/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D5/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D5/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D5/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D5/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D60/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D60/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D60/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{D60/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{ev}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E8	$B5_{ev}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D0/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D0/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D0/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D0/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D1/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D1/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D1/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D1/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D5/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D5/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D5/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D5/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D60/1}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D60/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D60/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{D60/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{ev}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	E11b	$B5_{ev}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D0/1}$	$B5_{60/10}$	NLX	1,687	[1,560, 1,819]	[1,579, 1,798]	<0.001	**W**
H1d	Mixed	$B5_{D0/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D0/10}$	$B5_{60/10}$	NLX	1,441	[1,304, 1,581]	[1,324, 1,558]	<0.001	**W**
H1d	Mixed	$B5_{D0/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D1/1}$	$B5_{60/10}$	NLX	1,607	[1,481, 1,733]	[1,500, 1,714]	<0.001	**W**

Hyp.	Scen.	Arm	Reference	Out.	$\bar{d}$	95% CI	90% CI	p_{Holm}	Class
H1d	Mixed	$B5_{D1/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D1/10}$	$B5_{60/10}$	NLX	1,353	[1,231, 1,478]	[1,252, 1,458]	<0.001	**W**
H1d	Mixed	$B5_{D1/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D5/1}$	$B5_{60/10}$	NLX	1,360	[1,245, 1,477]	[1,263, 1,459]	<0.001	**W**
H1d	Mixed	$B5_{D5/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D5/10}$	$B5_{60/10}$	NLX	1,149	[1,032, 1,268]	[1,052, 1,250]	<0.001	**W**
H1d	Mixed	$B5_{D5/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D60/1}$	$B5_{60/10}$	NLX	38.1	[30.5, 46.3]	[31.7, 44.9]	0.005	eq^{s}
H1d	Mixed	$B5_{D60/1}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D60/10}$	$B5_{60/10}$	NLX	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{D60/10}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id
H1d	Mixed	$B5_{ev}$	$B5_{60/10}$	NLX	1,716	[1,589, 1,844]	[1,607, 1,825]	<0.001	**W**
H1d	Mixed	$B5_{ev}$	$B5_{60/10}$	O1	0	[0, 0]	[0, 0]	1.000	id

Selective-hold grid. 2940 runs completed with 0 errors (21 grid points); the tracer matched in 2940 runs and 294 of 294 audited runs passed.

Table S25. Selective-hold grid (seeds 21–40; 120 s production, about 2,880 legitimate observations per run): means per grid point and arm. eq: equivocation; supp.: suppressed messages; delay: long-delayed messages; long supp.: suppression runs of 20 messages.

Point	Arm	O1	O2	NLX (rec-s)	W (s)	AV5
eq 5%, delay 1%	PBTI	0.00	19.4	0	1.67	0.997
	PBTI-SH	0.00	23.4	0	2.05	0.993
	PBTI-SH-T30	0.00	19.4	0	1.84	0.993
	$B5_{60/10}$	0.00	19.4	0	10.00	0.867
	$B5_{ev}$	0.00	19.4	0	1.67	0.996
	B5S	0.00	389.1	0	31.04	0.867
	$\text{PBTI-R}_{60/10}$	0.00	19.4	0	1.67	0.997
eq 5%, delay 5%	PBTI	0.00	19.8	0	1.78	0.996
	PBTI-SH	0.00	23.8	0	2.16	0.992
	PBTI-SH-T30	0.00	19.8	0	1.96	0.992
	$B5_{60/10}$	0.00	19.9	0	9.89	0.871
	$B5_{ev}$	0.00	19.8	0	1.79	0.996
	B5S	0.00	377.7	0	30.16	0.871
	$\text{PBTI-R}_{60/10}$	0.00	19.8	0	1.78	0.996
eq 5%, delay 10%	PBTI	0.00	20.4	0	1.95	0.996
	PBTI-SH	0.00	24.4	0	2.34	0.990
	PBTI-SH-T30	0.00	20.4	0	2.13	0.990
	$B5_{60/10}$	0.00	20.6	0	10.37	0.867
	$B5_{ev}$	0.00	20.4	0	1.96	0.995
	B5S	0.00	387.6	0	31.09	0.867
	$\text{PBTI-R}_{60/10}$	0.00	20.4	0	1.95	0.996
eq 0%	PBTI	0.00	0.0	0	0.03	1.000
	PBTI-SH	0.00	0.0	0	0.03	1.000
	PBTI-SH-T30	0.00	0.0	0	0.03	1.000
	$B5_{60/10}$	0.00	0.0	0	0.03	1.000
	$B5_{ev}$	0.00	0.0	0	0.03	1.000
	B5S	0.00	0.0	0	0.03	1.000
	$\text{PBTI-R}_{60/10}$	0.00	0.0	0	0.03	1.000
eq 0%, supp. 1%	PBTI	0.00	28.1	0	2.30	0.990
	PBTI-SH	0.00	28.1	0	2.30	0.990
	PBTI-SH-T30	0.00	28.1	0	2.30	0.990
	$B5_{60/10}$	0.00	28.1	0	2.30	0.990
	$B5_{ev}$	0.00	28.1	0	2.30	0.990
	B5S	0.00	28.1	0	2.30	0.990
	$\text{PBTI-R}_{60/10}$	0.00	28.1	0	2.30	0.990
eq 0%, supp. 5%	PBTI	0.00	138.2	0	11.46	0.952
	PBTI-SH	0.00	138.2	0	11.46	0.952
	PBTI-SH-T30	0.00	138.2	0	11.46	0.952

Point	Arm	O1	O2	NLX (rec-s)	*W* (s)	AV5
	$B5_{60/10}$	0.00	138.2	0	11.46	0.952
	$B5_{ev}$	0.00	138.2	0	11.46	0.952
	B5S	0.00	138.2	0	11.46	0.952
	PBTI-$R_{60/10}$	0.00	138.2	0	11.46	0.952
eq 0%, supp. 10%	PBTI	0.00	257.9	0	21.39	0.910
	PBTI-SH	0.00	257.9	0	21.39	0.910
	PBTI-SH-T30	0.00	257.9	0	21.39	0.910
	$B5_{60/10}$	0.00	257.9	0	21.39	0.910
	$B5_{ev}$	0.00	257.9	0	21.39	0.910
	B5S	0.00	257.9	0	21.39	0.910
	PBTI-$R_{60/10}$	0.00	257.9	0	21.39	0.910
eq 1%	PBTI	0.00	4.3	0	0.38	0.999
	PBTI-SH	0.00	7.1	0	0.62	0.998
	PBTI-SH-T30	0.00	4.3	0	0.47	0.998
	$B5_{60/10}$	0.00	4.3	0	4.37	0.937
	$B5_{ev}$	0.00	4.3	0	0.38	0.999
	B5S	0.00	181.7	0	13.88	0.937
	PBTI-$R_{60/10}$	0.00	4.3	0	0.38	0.999
eq 1%, supp. 1%	PBTI	0.00	32.4	0	2.66	0.990
	PBTI-SH	0.00	35.9	0	2.95	0.988
	PBTI-SH-T30	0.00	32.4	0	2.75	0.988
	$B5_{60/10}$	0.00	32.4	0	6.59	0.929
	$B5_{ev}$	0.00	32.4	0	2.66	0.990
	B5S	0.00	207.3	0	15.97	0.929
	PBTI-$R_{60/10}$	0.00	32.4	0	2.66	0.990
eq 1%, supp. 5%	PBTI	0.25	142.3	64	11.78	0.951
	PBTI-SH	0.15	147.7	39	12.22	0.950
	PBTI-SH-T30	0.25	142.3	61	11.90	0.950
	$B5_{60/10}$	0.25	142.6	64	15.39	0.896
	$B5_{ev}$	0.25	142.4	64	11.80	0.951
	B5S	0.25	301.9	64	23.89	0.896
	PBTI-$R_{60/10}$	0.25	142.3	64	11.78	0.951
eq 1%, supp. 10%	PBTI	0.25	261.8	53	21.71	0.910
	PBTI-SH	0.10	269.6	24	22.33	0.908
	PBTI-SH-T30	0.25	261.8	49	21.85	0.908
	$B5_{60/10}$	0.15	261.9	34	25.26	0.855
	$B5_{ev}$	0.20	261.8	44	21.71	0.910
	B5S	0.15	419.1	34	33.66	0.855
	PBTI-$R_{60/10}$	0.25	261.8	53	21.71	0.910
eq 5%	PBTI	0.00	19.1	0	1.63	0.996
	PBTI-SH	0.00	23.1	0	2.01	0.993
	PBTI-SH-T30	0.00	19.1	0	1.80	0.993
	$B5_{60/10}$	0.00	19.1	0	9.91	0.867
	$B5_{ev}$	0.00	19.1	0	1.64	0.996
	B5S	0.00	386.4	0	30.82	0.867
	PBTI-$R_{60/10}$	0.00	19.1	0	1.63	0.996
eq 5%, supp. 1%	PBTI	0.05	47.1	11	3.90	0.987
	PBTI-SH	0.00	52.0	0	4.35	0.983
	PBTI-SH-T30	0.05	47.1	9	4.08	0.983
	$B5_{60/10}$	0.05	47.1	8	12.07	0.858
	$B5_{ev}$	0.05	47.1	11	3.91	0.987
	B5S	0.00	409.8	0	32.73	0.858
	PBTI-$R_{60/10}$	0.05	47.1	11	3.90	0.987
eq 5%, supp. 5%	PBTI	0.65	156.6	162	12.99	0.949
	PBTI-SH	0.00	164.8	5	13.71	0.944
	PBTI-SH-T30	0.65	156.6	142	13.20	0.944
	$B5_{60/10}$	0.45	156.9	117	20.77	0.827
	$B5_{ev}$	0.60	156.7	151	13.00	0.948
	B5S	0.40	501.4	109	40.34	0.827
	PBTI-$R_{60/10}$	0.65	156.6	162	12.99	0.949
eq 5%, supp. 10%	PBTI	2.00	274.6	487	22.80	0.907
	PBTI-SH	0.00	285.8	19	23.75	0.902

Point	Arm	O1	O2	NLX (rec-s)	*W* (s)	AV5
	PBTI-SH-T30	2.00	274.6	426	23.02	0.902
	$B5_{60/10}$	0.90	275.2	208	29.96	0.795
	$B5_{ev}$	1.60	275.0	395	22.83	0.907
	B5S	0.50	591.5	141	47.82	0.795
	PBTI-$R_{60/10}$	2.00	274.6	487	22.80	0.907
eq 5%, long supp. 1%	PBTI	0.20	42.5	46	3.46	0.988
	PBTI-SH	0.00	46.6	0	3.84	0.984
	PBTI-SH-T30	0.20	42.5	39	3.63	0.984
	$B5_{60/10}$	0.00	42.5	0	11.69	0.859
	$B5_{ev}$	0.20	42.5	46	3.47	0.988
	B5S	0.00	408.0	0	32.51	0.859
	PBTI-$R_{60/10}$	0.20	42.5	46	3.46	0.988
eq 5%, long supp. 5%	PBTI	1.85	160.1	430	12.97	0.947
	PBTI-SH	0.00	165.5	20	13.45	0.943
	PBTI-SH-T30	1.85	160.1	373	13.14	0.943
	$B5_{60/10}$	0.55	160.2	145	20.41	0.831
	$B5_{ev}$	1.50	160.2	354	12.98	0.947
	B5S	0.50	490.0	138	39.15	0.831
	PBTI-$R_{60/10}$	1.85	160.1	430	12.97	0.947
eq 10%	PBTI	0.00	41.0	0	3.45	0.993
	PBTI-SH	0.00	45.0	0	3.83	0.985
	PBTI-SH-T30	0.00	41.0	0	3.62	0.985
	$B5_{60/10}$	0.00	41.0	0	11.98	0.855
	$B5_{ev}$	0.00	41.0	0	3.46	0.992
	B5S	0.00	419.7	0	33.96	0.855
	PBTI-$R_{60/10}$	0.00	41.0	0	3.45	0.993
eq 10%, supp. 1%	PBTI	0.25	68.8	56	5.70	0.983
	PBTI-SH	0.00	73.7	0	6.15	0.975
	PBTI-SH-T30	0.25	68.8	48	5.88	0.975
	$B5_{60/10}$	0.00	68.8	0	14.13	0.847
	$B5_{ev}$	0.15	68.8	36	5.71	0.982
	B5S	0.00	442.9	0	35.85	0.847
	PBTI-$R_{60/10}$	0.25	68.8	56	5.70	0.983
eq 10%, supp. 5%	PBTI	1.20	177.5	295	14.72	0.945
	PBTI-SH	0.00	185.7	6	15.45	0.937
	PBTI-SH-T30	1.20	177.5	258	14.93	0.937
	$B5_{60/10}$	0.40	178.1	100	22.74	0.816
	$B5_{ev}$	1.00	178.0	253	14.77	0.944
	B5S	0.30	532.0	85	43.21	0.816
	PBTI-$R_{60/10}$	1.20	177.5	295	14.72	0.945
eq 10%, supp. 10%	PBTI	2.95	294.9	724	24.48	0.904
	PBTI-SH	0.00	305.9	7	25.43	0.895
	PBTI-SH-T30	2.95	294.9	633	24.71	0.895
	$B5_{60/10}$	0.95	295.9	207	32.15	0.781
	$B5_{ev}$	2.10	295.6	513	24.54	0.903
	B5S	0.45	633.1	127	51.63	0.781
	PBTI-$R_{60/10}$	2.95	294.9	724	24.48	0.904

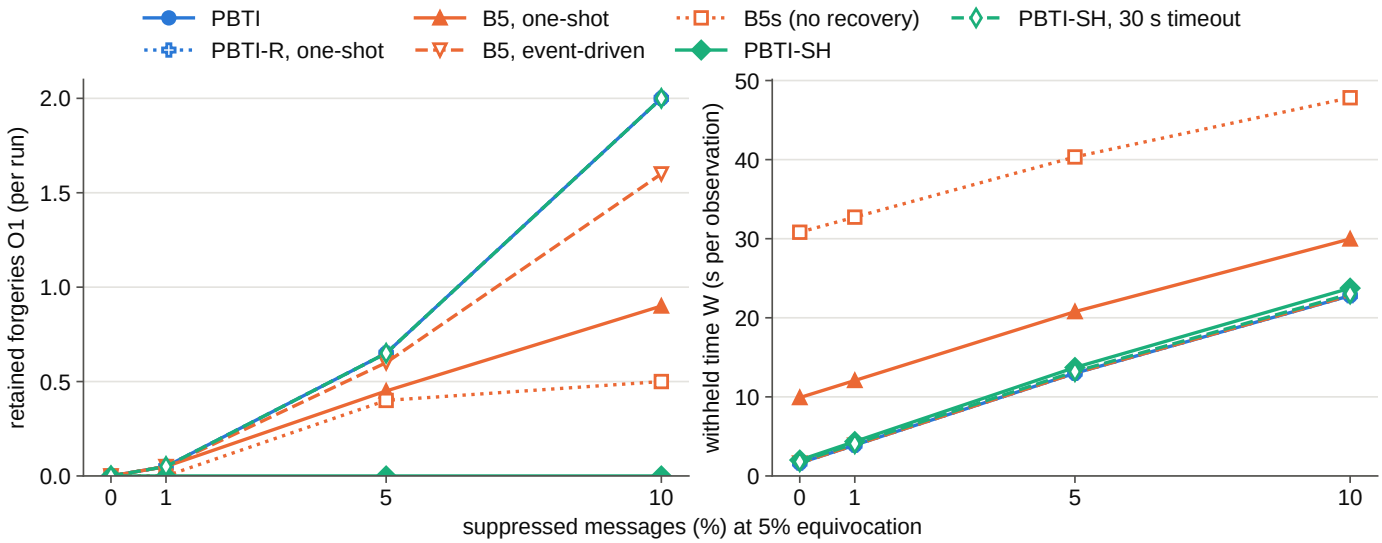


Figure S2. Selective-hold grid at 5% equivocation: retained forgeries O1 (left) and withheld time W (right) versus the rate of suppressed messages (means of 20 paired seeds, seeds 21–40; 120 s production). One-shot: one-shot review rule with $R = 60$ s and $C = 10$ s (PBTI-R$_{60/10}$ and B5$_{60/10}$); event-driven: B5$_{ev}$. In the left panel PBTI, PBTI-R$_{60/10}$ and PBTI-SH-T30 coincide. Colour marks the pipeline family, line style and marker the variant. All points: Table S25.

Selective-hold factorial cells (amendment-informed). Added after the freeze and before any confirmatory selective-hold run, because the pilot (seed 101) showed retained forgeries at only one of the 21 points of the registered grid; with 20 seeds, the grid later showed small numbers at ten points (Table S25). The amendment added the eight factorial cells with equivocation high (10%), four of them with suppressed runs high (FG), on fresh seeds 21–40. 1120 runs, 0 errors; the tracer matched in 1120 runs and 112 of 112 audited runs passed. Gates: G3F holds, G4F holds.

All confirmatory comparisons, with mean differences, 95% and 90% bootstrap intervals, raw and Holm-adjusted p and classes, are in Data S1; the comparisons on which gates G3, G4, G3F and G4F depend follow. The class "eq" of the grid row with 5% equivocation and 5% suppressed messages (O1, versus PBTI) is resampling-dependent: its 90% interval touches -1, the margin, and the comparison of the same data with PBTI-R$_{60/10}$ is classed inconclusive in Data S1; no gate depends on it. "dw" (different within the margin) means significant with a 95% interval overlapping the margin, so the point estimate may exceed the margin.

Table S26. Selective holds in the factorial cells (means over seeds 21–40; 120 s production, about 2,880 legitimate observations per run). FG: equivocation and suppressed runs both high.

Cell (high factors)	Arm	O1	O2	NLX (rec-s)	*W* (s)	AV5
run2	PBTI	0.00	251.8	0	21.69	0.919
(duplicates, clock error,	PBTI-SH	0.00	286.4	0	24.69	0.904
equivocation)	PBTI-SH-T30	0.00	251.8	0	22.17	0.904
	$B5_{60/10}$	0.00	256.3	0	29.78	0.793
	$B5_{ev}$	0.00	255.3	0	22.00	0.917
	B5S	0.00	598.2	0	49.46	0.793
	PBTI-$R_{60/10}$	0.00	251.8	0	21.69	0.919
run3^{FG}	PBTI	1.40	243.0	325	20.46	0.921
(loss, duplicates, equivocation,	PBTI-SH	0.00	256.1	0	21.60	0.910
suppressed runs)	PBTI-SH-T30	1.40	243.0	281	20.73	0.910
	$B5_{60/10}$	0.40	246.0	62	28.72	0.791
	$B5_{ev}$	1.15	244.4	267	20.58	0.920
	B5S	0.00	602.5	0	49.51	0.791
	PBTI-$R_{60/10}$	1.40	243.0	325	20.46	0.921
run4^{FG}	PBTI	2.50	478.7	613	40.54	0.841
(reordering, clock error,	PBTI-SH	0.00	516.2	6	43.74	0.824
equivocation, suppressed runs)	PBTI-SH-T30	2.50	478.7	536	41.04	0.824
	$B5_{60/10}$	0.85	484.1	185	47.86	0.727
	$B5_{ev}$	1.50	482.3	383	40.85	0.838
	B5S	0.35	789.9	99	65.46	0.727
	PBTI-$R_{60/10}$	2.50	478.7	613	40.54	0.841
run5	PBTI	0.00	43.0	0	3.97	0.991
(loss, reordering, equivocation)	PBTI-SH	0.00	47.0	0	4.37	0.982
	PBTI-SH-T30	0.00	43.0	0	4.16	0.982
	$B5_{60/10}$	0.00	43.4	0	12.59	0.852
	$B5_{ev}$	0.00	43.0	0	3.98	0.991
	B5S	0.00	424.6	0	34.78	0.852
	PBTI-$R_{60/10}$	0.00	43.0	0	3.97	0.991
run8^{FG}	PBTI	1.65	270.6	405	22.69	0.912
(long delay, equivocation,	PBTI-SH	0.00	282.7	2	23.76	0.902
suppressed runs)	PBTI-SH-T30	1.65	270.6	354	22.95	0.902
	$B5_{60/10}$	0.50	273.9	96	30.99	0.785
	$B5_{ev}$	0.80	272.5	199	22.86	0.911
	B5S	0.15	625.8	43	51.26	0.785
	PBTI-$R_{60/10}$	1.65	270.6	405	22.69	0.912
run9	PBTI	0.00	241.0	0	21.42	0.913
(loss, long delay, clock error,	PBTI-SH	0.00	272.9	0	24.25	0.895
equivocation)	PBTI-SH-T30	0.00	241.0	0	21.92	0.895
	$B5_{60/10}$	0.00	245.5	0	29.67	0.787
	$B5_{ev}$	0.00	243.6	0	21.66	0.911
	B5S	0.00	592.9	0	49.61	0.787
	PBTI-$R_{60/10}$	0.00	241.0	0	21.42	0.913
run14	PBTI	0.00	45.0	0	4.11	0.991
(duplicates, reordering,	PBTI-SH	0.00	49.0	0	4.51	0.982
long delay, equivocation)	PBTI-SH-T30	0.00	45.0	0	4.31	0.982
	$B5_{60/10}$	0.00	45.4	0	12.97	0.853
	$B5_{ev}$	0.00	45.0	0	4.13	0.990
	B5S	0.00	431.3	0	35.23	0.853
	PBTI-$R_{60/10}$	0.00	45.0	0	4.11	0.991
run15^{FG}	PBTI	0.85	403.7	195	35.08	0.853
(loss, duplicates, reordering,	PBTI-SH	0.00	444.4	0	38.63	0.832
long delay, clock error,	PBTI-SH-T30	0.85	403.7	167	35.67	0.832
equivocation, suppressed runs)	$B5_{60/10}$	0.25	414.1	42	43.32	0.735
	$B5_{ev}$	0.50	410.1	117	35.62	0.850
	B5S	0.00	737.5	0	61.83	0.735
	PBTI-$R_{60/10}$	0.85	403.7	195	35.08	0.853

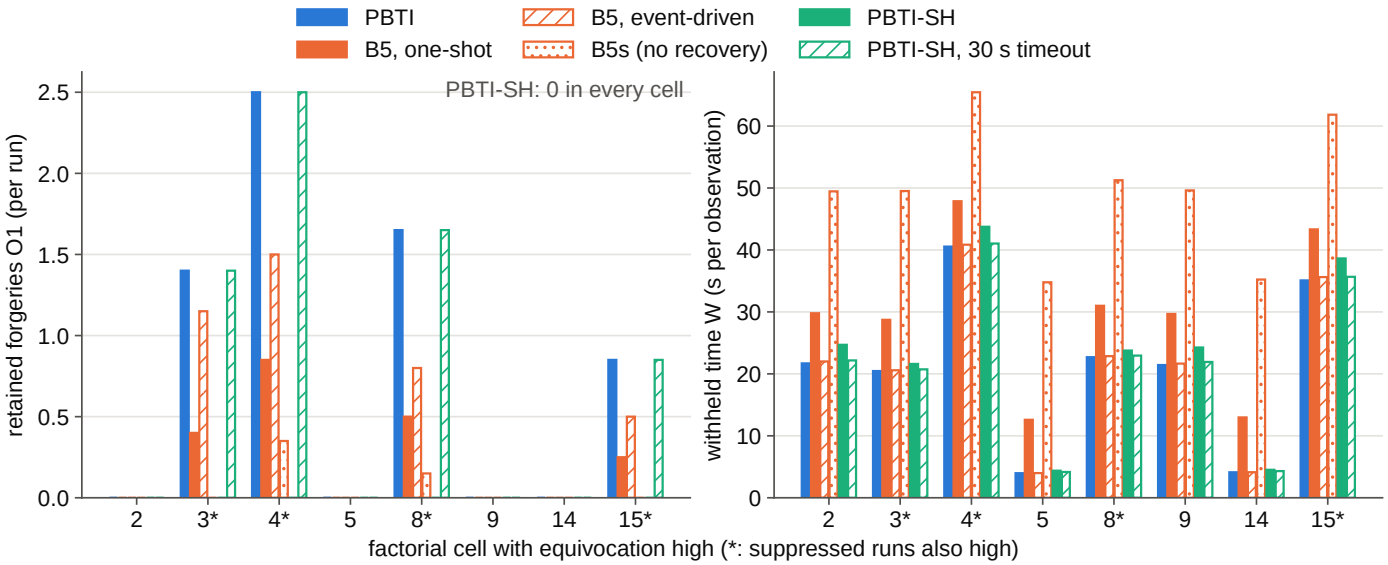


Figure S3. Selective holds in the eight factorial cells with equivocation high (amendment-informed; means of 20 paired seeds, 120 s production, about 2880 legitimate observations per run): retained forgeries O1 (left) and withheld time W (right). *: suppressed runs also high. Colour marks the pipeline family (PBTI blue, B5 orange, PBTI-SH aqua) and hatching the variant (one-shot: $B5_{60/10}$; event-driven: $B5_{ev}$); PBTI-R is identical to PBTI in every cell and not shown. Values: Table S26.

Table S27. Comparisons that decide the selective-hold gates: PBTI-SH versus the reference at the joint-grid points at which PBTI retained forgeries (G3: O1 or NLX better beyond the margin at half of these points; G4: G3 and W not worse than $B5_{60/10}$ there) and in the factorial cells with equivocation and suppressed runs high (amendment 1; G3F: O1 or NLX better beyond the margin in at least two of the four cells; G4F: G3F and W not worse than $B5_{60/10}$). Means over seeds 21–40; $\bar{d}$ = PBTI-SH − reference; Holm within campaign and outcome. Classes: id, identical in every seed; eq, equivalent within the margin (90% CI inside ±margin); eq^s, equivalent and significant after Holm correction; **B**/**W**, better/worse beyond the margin (Holm-adjusted $p < 0.05$ and 95% CI beyond the margin); dw, significant, margin unresolved (Holm-adjusted $p < 0.05$, 95% CI crossing a margin boundary; historical label "different within the margin"); inc, inconclusive. All comparisons: Data S1.

Campaign	Point or cell	Ref.	Out.	SH	Ref.	$\bar{d}$	95% CI	p_{Holm}	Class
grid	eq 1%, supp. 5%	PBTI	O1	0.15	0.25	−0.1	[−0.25, 0]	1.000	eq
		PBTI	NLX	39	64	−25.1	[−62.7, 0]	1.000	eq
		$B5_{60/10}$	W	12.22	15.39	−3.17	[−3.77, −2.54]	<0.001	**B**
grid	eq 1%, supp. 10%	PBTI	O1	0.10	0.25	−0.15	[−0.3, 0]	1.000	eq
		PBTI	NLX	24	53	−29	[−59, 0]	1.000	eq
		$B5_{60/10}$	W	22.33	25.26	−2.93	[−3.49, −2.38]	<0.001	**B**
grid	eq 5%, supp. 1%	PBTI	O1	0.00	0.05	−0.05	[−0.15, 0]	1.000	eq
		PBTI	NLX	0	11	−11	[−33, 0]	1.000	eq
		$B5_{60/10}$	W	4.35	12.07	−7.72	[−7.98, −7.46]	<0.001	**B**
grid	eq 5%, supp. 5%	PBTI	O1	0.00	0.65	−0.65	[−1.05, −0.3]	0.618	eq
		PBTI	NLX	5	162	−157	[−250, −74.2]	0.830	inc
		$B5_{60/10}$	W	13.71	20.77	−7.06	[−7.3, −6.82]	<0.001	**B**
grid	eq 5%, supp. 10%	PBTI	O1	0.00	2.00	−2	[−2.5, −1.45]	0.021	**B**
		PBTI	NLX	19	487	−468	[−595, −349]	0.024	**B**
		$B5_{60/10}$	W	23.75	29.96	−6.22	[−6.55, −5.89]	<0.001	**B**
grid	eq 5%, long supp. 1%	PBTI	O1	0.00	0.20	−0.2	[−0.55, 0]	1.000	eq
		PBTI	NLX	0	46	−45.5	[−126, 0]	1.000	inc
		$B5_{60/10}$	W	3.84	11.69	−7.85	[−8.11, −7.59]	<0.001	**B**
grid	eq 5%, long supp. 5%	PBTI	O1	0.00	1.85	−1.85	[−2.65, −1.1]	0.159	inc
		PBTI	NLX	20	430	−409	[−586, −241]	0.168	inc
		$B5_{60/10}$	W	13.45	20.41	−6.95	[−7.42, −6.46]	<0.001	**B**
grid	eq 10%, supp. 1%	PBTI	O1	0.00	0.25	−0.25	[−0.5, −0.05]	1.000	eq

Campaign	Point or cell	Ref.	Out.	SH	Ref.	$\bar{d}$	95% CI	p_{Holm}	Class
		PBTI	NLX	0	56	−56.2	[−113, −11.4]	1.000	inc
		$B5_{60/10}$	W	6.15	14.13	−7.98	[−8.15, −7.79]	<0.001	**B**
grid	eq 10%, supp. 5%	PBTI	O1	0.00	1.20	−1.2	[−1.6, −0.85]	0.034	dw
		PBTI	NLX	6	295	−290	[−378, −202]	0.035	**B**
		$B5_{60/10}$	W	15.45	22.74	−7.29	[−7.51, −7.09]	<0.001	**B**
grid	eq 10%, supp. 10%	PBTI	O1	0.00	2.95	−2.95	[−3.75, −2.2]	0.015	**B**
		PBTI	NLX	7	724	−717	[−908, −528]	<0.001	**B**
		$B5_{60/10}$	W	25.43	32.15	−6.72	[−6.99, −6.44]	<0.001	**B**
cells	$run3^{FG}$	PBTI	O1	0.00	1.40	−1.4	[−1.8, −0.95]	0.016	dw
		PBTI	NLX	0	325	−325	[−419, −230]	0.009	**B**
		PBTI	W	21.60	20.46	1.14	[1.02, 1.27]	<0.001	**W**
		PBTI	O2	256.1	243.0	13.1	[11.6, 14.7]	0.003	eq^s
		PBTI	AV5	0.910	0.921	−0.011	[−0.0119, −0.0102]	0.004	**W**
		$B5_{60/10}$	W	21.60	28.72	−7.12	[−7.33, −6.92]	<0.001	**B**
		$B5_{60/10}$	O2	256.1	246.0	10.1	[9, 11.2]	0.003	eq^s
cells	$run4^{FG}$	PBTI	O1	0.00	2.50	−2.5	[−3.2, −1.9]	0.006	**B**
		PBTI	NLX	6	613	−607	[−781, −455]	<0.001	**B**
		PBTI	W	43.74	40.54	3.2	[3.09, 3.32]	<0.001	**W**
		PBTI	O2	516.2	478.7	37.5	[36.1, 38.9]	0.003	**W**
		PBTI	AV5	0.824	0.841	−0.0161	[−0.0169, −0.0152]	0.004	**W**
		$B5_{60/10}$	W	43.74	47.86	−4.12	[−4.28, −3.95]	<0.001	**B**
		$B5_{60/10}$	O2	516.2	484.1	32	[31.1, 33.1]	0.003	**W**
cells	$run8^{FG}$	PBTI	O1	0.00	1.65	−1.65	[−2.5, −0.95]	0.021	dw
		PBTI	NLX	2	405	−403	[−589, −242]	0.009	**B**
		PBTI	W	23.76	22.69	1.07	[0.944, 1.2]	<0.001	**W**
		PBTI	O2	282.7	270.6	12.2	[10.6, 13.7]	0.003	eq^s
		PBTI	AV5	0.902	0.912	−0.0104	[−0.0112, −0.00964]	0.004	dw
		$B5_{60/10}$	W	23.76	30.99	−7.23	[−7.47, −6.99]	<0.001	**B**
		$B5_{60/10}$	O2	282.7	273.9	8.8	[7.9, 9.75]	0.003	eq^s
cells	$run15^{FG}$	PBTI	O1	0.00	0.85	−0.85	[−1.3, −0.45]	0.092	inc
		PBTI	NLX	0	195	−194	[−300, −102]	0.024	**B**
		PBTI	W	38.63	35.08	3.55	[3.35, 3.74]	<0.001	**W**
		PBTI	O2	444.4	403.7	40.7	[38.2, 43.1]	0.003	**W**
		PBTI	AV5	0.832	0.853	−0.0211	[−0.0222, −0.02]	0.004	**W**
		$B5_{60/10}$	W	38.63	43.32	−4.69	[−5.03, −4.33]	<0.001	**B**
		$B5_{60/10}$	O2	444.4	414.1	30.4	[28.2, 32.4]	0.003	dw

Table S28. Selective-hold controls: PBTI (re-executed main-evaluation rows) and PBTI-SH on the 16 workloads, and PBTI-SH-T30 on four of them (means over seeds 1–20; descriptive).

	O1		O2		NLX (rec-s)		*W* (s)		AV5	
Scenario	PBTI	SH	PBTI	SH	PBTI	SH	PBTI	SH	PBTI	SH
E1	0.0	0.0	0.0	0.0	0	0	0.03	0.03	1.000	1.000
E2	0.0	0.0	354.9	354.9	0	0	16.28	16.28	0.951	0.951
E3	0.0	0.0	427.7	427.7	0	0	19.55	19.55	0.941	0.941
E4	0.0	0.0	0.0	0.0	0	0	0.03	0.03	1.000	1.000
E5	150.0	150.0	222.6	225.6	38,266	38,242	8.64	8.81	0.972	0.969
E6	0.0	0.0	0.0	0.0	0	0	0.03	0.03	1.000	1.000
E7	0.0	0.0	0.0	0.0	0	0	0.07	0.07	1.000	1.000
E8	0.0	0.0	12.0	14.0	0	0	0.56	0.67	0.999	0.999
E9	0.0	0.0	0.0	0.0	0	0	0.47	0.47	1.000	1.000
E10	0.0	0.0	0.0	0.0	0	0	0.07	0.07	1.000	1.000
E11/12	0.0	0.0	0.0	0.0	0	0	6.76	6.76	0.889	0.889
E11b	0.0	0.0	193.3	193.3	0	0	17.54	17.54	0.867	0.867
E13	0.0	0.0	0.0	0.0	0	0	6.59	6.59	0.894	0.894
E14	0.0	0.0	270.1	270.1	8,042	8,042	11.41	11.41	0.974	0.974
E15	15.8	15.8	89.5	89.5	5,145	5,145	4.11	4.11	0.988	0.988
Mixed	5.2	5.0	335.0	361.6	3,462	3,284	22.95	24.17	0.847	0.841
SH = PBTI-SH-T30 (30 s timeout):										
E5	150.0	150.0	222.6	222.6	38,266	38,242	8.64	8.75	0.972	0.969
E8	0.0	0.0	12.0	12.0	0	0	0.56	0.63	0.999	0.999
E14	0.0	0.0	270.1	270.1	8,042	8,042	11.41	11.41	0.974	0.974
Mixed	5.2	5.2	335.0	335.0	3,462	3,341	22.95	23.18	0.847	0.841

S6. Revocation, Sensitivity Analyses, Replay, Ablation and Insider Tampering

Table S29. Lifecycle analysis of reviewed release under revoked credentials (deterministic; repository `scripts/a5_lifecycle_matrix.py`): device and relay keys compromised at t_c = 10 s, revocation observed at 20 s with t_{comp} = 10 s, reviewer decisions at 30 s (in the superseded-snapshot case a decision at 19.5 s bound to a superseded snapshot is refused first), ticks to 60 s. Release-evidence policies of the REVOKED_BUFFERED waiver: none; discretionary (the observation's own time stamp precedes t_{comp}; used in all scenario campaigns); witness (a token from a time-stamping witness for the exact statement digest, witness time plus clock uncertainty (1 s) before t_{low}). PBTI and B5 gave identical results. The interval row sets $t_{\text{low}} = t_c -$ width/2. The witness's key and clock are trusted by assumption.

Case (PBTI; B5 identical)	none	discretionary	witness
Legitimate buffered record, witnessed before t_c	withheld (rejected)	recovered	recovered
Backdated forgery, witness sees it after t_c	excluded (rejected)	**forgery authoritative**	excluded (rejected)
Backdated forgery, never witnessed	excluded (rejected)	**forgery authoritative**	excluded (rejected)
Forgery contradicted by an admitted successor	excluded (rejected)	excluded (rejected)	excluded (rejected)
Legitimate record, witness outage	withheld (rejected)	recovered	withheld (rejected)
Decision on a superseded snapshot (refused at 19.5 s; released by a fresh decision at 30 s)	recovered	recovered	recovered
Reviewer key revoked	withheld (quarantined)	withheld (quarantined)	withheld (quarantined)
STALE_TS case, origin revoked before release	withheld (rejected)	withheld (rejected)	withheld (rejected)
Released, later revocation covers commit time	released, then tainted	released, then tainted	released, then tainted
Released under revoked key, then earlier t_{comp} revision	withheld (rejected)	released, then tainted; window corrected	released, then tainted; window corrected
Interval width $t_{high} - t_{low}$ (witness policy)	0 s: recovered; 5 s: recovered; 30 s: withheld; 120 s: withheld		

Revocation. With immediate status refresh ($\Delta = 0$), PBTI tainted 279.9 commits per run in E14; with a 30 s refresh, 3 post-revocation injections per run passed before the refresh and were tainted once the revocation was observed.

Table S30. Revocation (E14): post-revocation admissions of attacks with revoked keys, and non-legitimate observations remaining authoritative at the end of the run, by status refresh interval Δ (means over 20 seeds).

	B1		B2		B3		PBTI	
Status refresh Δ	post-rev.	non-legit.	post-rev.	non-legit.	post-rev.	non-legit.	post-rev.	non-legit.
0 s	40	239.9	0	199.9	0	199.9	0	0
1 s	40	239.9	0	199.9	0	199.9	0	0
5 s	40	239.9	0	199.9	0	199.9	0	0
30 s	40	239.9	3	202.9	3	202.9	3	0

Review-workflow tests. The workflow is covered by unit tests: release of a stale quarantined observation by an authorized reviewer with a signed evidence record; refusal of an unauthorized reviewer and of a decision with an invalid signature; idempotent duplicate decision, unknown case and revoked reviewer; stale review and conflicting evidence (release fails, slot contested); rejection of an uncorroborated forgery; re-taint of a released observation after a later revocation with recomputation of the current value; recovery after a long gap with a five-second gap horizon; B5 feed hold and release; B5s frozen feed with a new epoch admitted; state determinism over 120 delivery permutations; 15 handcrafted traces of the time-dependent outcomes; the deferring and event-driven review rules and the selective-hold gate; and the witness tokens, lifecycle cases and the complete revocation–quarantine–review–release–taint–correction sequence.

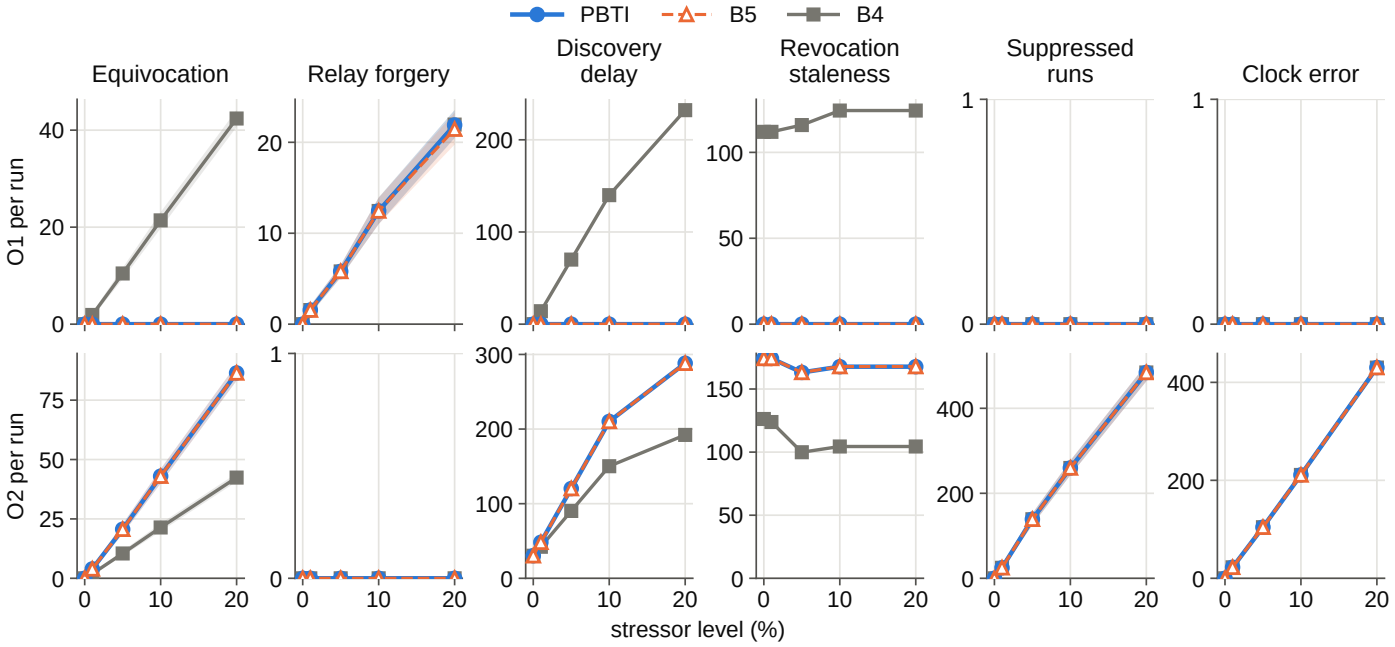


Figure S4. Stressor sweep: non-legitimate observations authoritative at the end of the run (O1, top) and legitimate observations missing (O2, bottom) per run versus stressor level for PBTI, B5 and B4 (means of 20 paired seeds, shaded 95% bootstrap intervals; 120 s production phase, 2880 legitimate observations per run; exploratory). Where B5 coincides with PBTI, its dashed line overlays PBTI's. For the five remaining factors (loss, duplicates, reordering, long delay and outage bursts, levels 0–20%), both outcomes were zero at every level for all three pipelines (stored summary in Data S1). Definitions of all factors and levels: repository `docs/V3_SWEEP_DESIGN.md`.

Table S31. Fractional factorial 2^{7-3}_{IV} at 0 and 10%: estimated effects on O1 and O2 per run (mean over 20 seeds of the per-seed effect; *: 95% t interval excludes 0; exploratory). A loss, B duplicates, C reordering, D long delay, E clock error ($= ABC$), F equivocation ($= BCD$), G suppressed runs ($= ACD$); main effects are aliased with three-factor interactions, two-factor interactions in the chains shown.

	PBTI		B4		B5	
Effect	O1	O2	O1	O2	O1	O2
A (loss)	-0.0	-10.0*	0.1	-9.9*	0.1	-9.4*
B (dup)	-0.0	-10.7*	-0.0	-10.7*	0.1	-10.1*
C (reorder)	0	-1.6*	-0.3	-2.7*	-0.0	-1.4
D (delay)	-0.1	-15.2*	0.5	-15.8*	-0.0	-14.6*
E (clock)	0	193.6*	-0.9*	194.5*	-0.0	195.9*
F (equivocation)	1.1*	42.1*	21.6*	20.5*	0.4*	46.2*
G (missing)	1.1*	210.6*	0.9*	211.4*	0.4*	212.1*
AB = CE = FG	1.1*	-1.3*	0.9*	-0.5	0.4*	0.1
AC = BE = DG	-0.1	-9.4*	0.3	-9.2*	-0.0	-9.0*
AD = EF = CG	0	-3.9*	-0.9*	-3.0*	-0.0	-1.6*
AE = BC = DF	-0.1	2.1*	0.5	1.5*	-0.0	2.7*
AF = DE = BG	-0.0	-17.5*	0.1	-17.4*	0.1	-16.9*
AG = CD = BF	-0.0	-9.8*	-0.0	-9.9*	0.1	-9.2*
BD = CF = EG	0	-7.4*	-0.3	-8.6*	-0.0	-7.2*

Table S32. Replay of SKAB telemetry with injected, labelled stressors (means over 20 seeds; about 14,400 legitimate observations per run; exploratory). Avail.: share of reachable legitimate observations authoritative at the end, in rows that SKAB labels normal or anomalous (process anomalies, not attacks). Unstressed controls: 27 runs with 0 missing and 0 non-legitimate observations in total. Attributed re-runs traced 52 of the 53 observations retained by PBTI and all 52 retained by B5 to a compromised, never-revoked gateway forging its legacy device's data, and 668 of B4's 1,073 to impersonation that B4 cannot taint.

Speed	Pipeline	Reach.	Non-legit.	Missing	Quar.	Held	Released (p50 s)	Avail. normal %	Avail. anomaly %
1×	B4	14,400	192.1	190.2	0	0	0	98.5	98.8
1×	B5	14,400	11.3	303.2	52.6	2572.2	2370.8 (65.2)	98.2	97.7
1×	PBTI	14,400	11.3	303.1	0	0	0	98.2	97.7
5×	B4	14,400	175.3	306.3	0	0	0	97.8	97.9
5×	B5	14,400	10.4	418.9	51.8	2567.2	2359.9 (65.3)	97.5	96.8
5×	PBTI	14,400	10.9	418.9	0	0	0	97.5	96.8
10×	B4	14,400	171.2	448.8	0	0	0	96.9	96.8
10×	B5	14,400	10.2	562.1	51.6	2571.8	2358.6 (65.9)	96.6	95.8
10×	PBTI	14,400	10.4	562.1	0	0	0	96.6	95.8

Table S33. Ablation study on the mixed workload (means over 20 seeds; baseline rows give B1–B4 on the same workload for reference; *: Holm-corrected Wilcoxon $p < 0.05$ versus full PBTI). Non-legit.: non-legitimate authoritative observations; Quar.: missing observations held in the quarantine archive; Ab1–Ab12: ablated variants.

Configuration	Non-legit.	Missing	Quar.	Dup.	CV wrong %	μs/msg
B1 (baseline)	485.5	282.6	0	0	28.1	51
B2 (baseline)	380.3	340.9	0	0	22.8	53
B3 (baseline)	215.1	272.6	0	0	6.8	239
B4 (baseline)	147.7	327.6	0	0	3.5	266
PBTI (full)	5.2	335	60.1	0	0.8	494
Ab1 no relay signature	5.9*	362.4*	60.6	0	3.2*	323*
Ab2 static policy	30.4*	482.3*	60.1	0	3.6*	478
Ab3 no origin signature	30.1*	335	60.1	0	21.0*	342*
Ab4 no prev link	14.3*	332.8*	60.1	0	0.8	484
Ab5 edge-assigned identity	72*	273*	60.0	48.1*	1.0*	495
Ab6 no conflict detection	40.7*	300.2*	60.1	0	1.0*	478
Ab7 no horizon/epoch checks	35.1*	274.9*	0*	0	4.6*	489
Ab8 arrival-order state	5.2	335	60.1	0	7.4*	481
Ab9 no evidence commitment	5.2	335	0*	0	0.8	424*
Ab10 no quarantine	5.2	335	0*	0	0.8	477
Ab11 no revocation/taint	45.5*	329.9*	60.1	0	0.8*	475
Ab12 volatile edge buffer	5.0	517.5*	60.6	0	0.8	498

Table S34. Detection of insider tampering by the offline auditor (runs with detection / runs; 50 objects altered per run, or one checkpoint deleted for T4). Re-run with the final auditor (70 runs): the same detection outcome in 70, PBTI detected 35 of 35, and the same check families fired (in T2 decision replay additionally reports disposition mismatches).

Insider action	B3 detected	PBTI detected	PBTI checks that fired
T1 raw object modified	5/5	5/5	A-DET, A-LIN, A-RAW
T2 raw object deleted	5/5	5/5	A-DET, A-LIN, A-RAW
T3 evidence record modified	5/5	5/5	A-DET, A-EVID, A-REG
T4 checkpoint deleted	0/5	5/5	A-EVID
T5 log rewritten + re-signed (key holder)	0/5	5/5	A-DET, A-EVID, A-REG
T6 derived record modified	0/5	5/5	A-LIN
T7 registry state modified	0/5	5/5	A-DET, A-LIN

S7. Performance and Deployment

Figure S5 shows the in-process batching and worker study, the per-stage cost of PBTI and the MQTT/TLS latency reported in the article's Section 6.8.

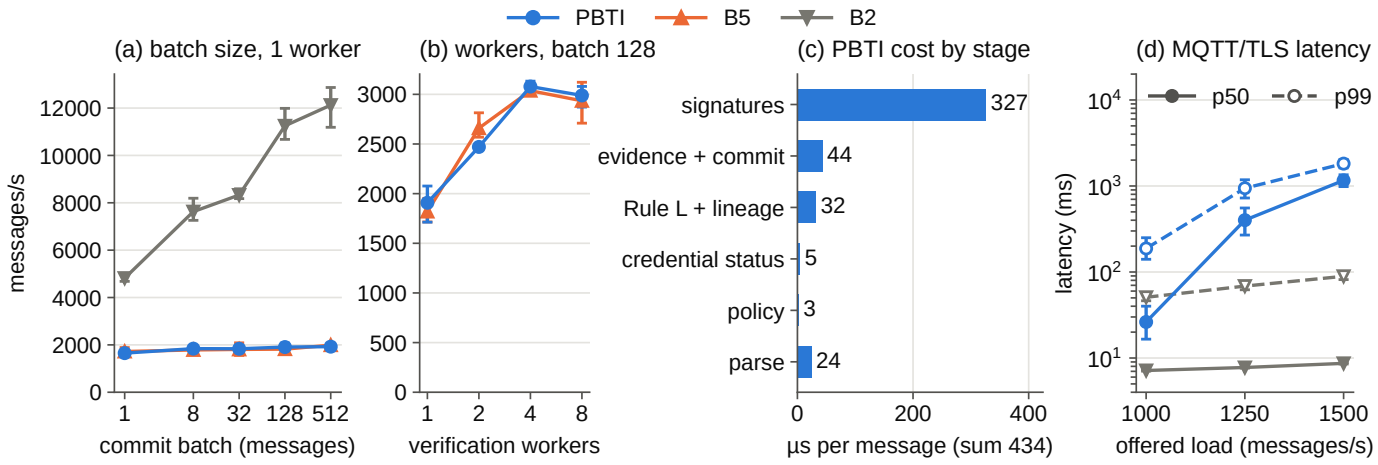


Figure S5. (**a**) In-process throughput versus group-commit batch size (one worker; mean and 95% CI, 3 repetitions). (**b**) PBTI and B5 versus signature-verification workers on two vCPUs (batch 128). (**c**) PBTI cost by stage (in-process benchmark; Rule L and lineage without the nested commit time, so that the stages add up to the per-message total). (**d**) MQTT/TLS latency on one host with all roles co-located, p50 (solid) and p99 (dashed), mean and 95% CI of 30 repetitions. Colours as in the article's Figure 2 (PBTI blue, B5 orange); B2 grey.

Containers and hardware protocols. A hardened container image (base pinned by digest, pinned Debian packages, hash-locked Python dependencies, non-root user, health check) built and passed the unit tests, a smoke run and an MQTT/TLS run, also with a read-only root file system and all capabilities dropped. The repository specifies a three-host MQTT/TLS deployment (edge publisher, broker, verifier; clock synchronization, RTT, bandwidth, 30 repetitions per cell) and a constrained-device signing benchmark (Raspberry-Pi-class; thermal state, power mode, energy). No separate hosts or constrained devices were available, so neither experiment was performed; no result in the article comes from them, and containers on one host are not reported as a multi-host deployment.

Reproduction. `python3 -m pytest -q tests/`; the campaign commands of Table S8; `python3 scripts/compare_a0.py results/v4/raw/A0_core.jsonl`; `python3 scripts/a5_lifecycle_matrix.py`; `python3 formal/v4_locality_lifecycle_check.py`; `python3 analysis/analyze_v4.py`; then the paper build. On the two-vCPU reference host the confirmatory and re-execution campaigns took about six hours; the minimal reproduction kit (`tools/reexec/reproduce_subset.sh`) took 81 minutes on two shared

vCPUs in the second re-execution, and 3 h 39 min with four parallel runs on the laptop of the author re-execution, and reports every difference.

Table S35. In-process verifier results with SQLite group commit (write-ahead logging, `synchronous=NORMAL`: a commit survives an application crash but can be rolled back by a power loss or operating-system crash; means over 10 repetitions; one two-vCPU host). With 10 repetitions per pipeline, normality cannot be established reliably (Shapiro–Wilk minimum $p = 0.05$), so non-parametric tests were used.

Pipeline	Throughput (msg/s) [95% CI]	CPU (μs/msg)	Service p50 (μs)	Service p99 (μs)	Wire (B/msg)	Stored (B/msg)	Edge CPU (μs/msg)
B1	11358 [11079, 11580]	73	33	139	219	674	38
B2	10707 [10410, 10946]	78	38	158	220	674	36
B3	3329 [3298, 3362]	281	222	581	292	1066	118
PBTI	2067 [2039, 2093]	466	402	1021	641	909	291

Table S36. Verifier saturation throughput (messages/s) by number of devices (in-process, SQLite group commit as in Table S35; means of 3 repetitions).

Pipeline	10 devices	100 devices	500 devices	2000 devices
B1	11583	10525	10924	10366
B2	10732	10012	10496	10043
B3	3360	3216	3142	3067
PBTI	2054	2064	2018	1945

Table S37. In-process batching and worker study (10,000 messages per run, 3 repetitions per cell, randomized order; 2 vCPUs). Latency: time from dispatch of a message's batch to its commit at saturation (the fill time of a batch at an open-loop arrival rate comes on top). Disposition sequences were identical across all cells of a pipeline.

Pipeline	Batch	Workers	msg/s [95% CI]	p50 (ms)	p99 (ms)	CPU cores	CPU μs/msg	DB B/msg
B2	1	1	4818 [4682, 4997]	0.1	1.0	0.64	133	673
B2	8	1	7627 [7259, 8192]	0.5	13.8	0.70	92	673
B2	32	1	8328 [8176, 8591]	2.3	19.9	0.75	90	673
B2	128	1	11250 [10677, 11988]	7.5	48.1	0.83	74	673
B2	512	1	12124 [11188, 12869]	32.8	86.7	0.90	75	673
B5	1	1	1720 [1584, 1888]	0.5	1.3	0.93	542	911
B5	8	1	1786 [1679, 1851]	3.7	17.3	0.94	526	911
B5	32	1	1808 [1555, 2087]	15.5	41.7	0.95	536	911
B5	128	1	1820 [1715, 1952]	65.0	122.2	0.97	533	911
B5	128	2	2659 [2568, 2814]	91.8	141.5	1.61	606	911
B5	128	4	3037 [3017, 3050]	160.5	229.8	1.79	591	911
B5	128	8	2935 [2710, 3122]	325.3	555.9	1.77	606	911
B5	512	1	1988 [1839, 2069]	237.2	374.0	0.98	494	911
PBTI	1	1	1651 [1584, 1708]	0.5	1.3	0.93	562	911
PBTI	1	2	862 [850, 878]	1.9	7.3	1.18	1366	911
PBTI	1	4	1113 [1051, 1145]	3.1	14.6	1.30	1166	911
PBTI	1	8	1255 [1229, 1272]	5.7	18.4	1.33	1,063	911
PBTI	8	1	1841 [1762, 1911]	3.5	16.6	0.95	514	911
PBTI	8	2	1618 [1543, 1674]	9.1	24.6	1.23	762	911
PBTI	8	4	1842 [1702, 2070]	16.2	40.3	1.42	774	911
PBTI	8	8	2056 [1935, 2170]	29.6	60.7	1.54	750	911
PBTI	32	1	1836 [1713, 1919]	14.9	37.3	0.95	521	911
PBTI	32	2	2037 [1981, 2143]	29.6	55.3	1.33	651	911
PBTI	32	4	2428 [2280, 2547]	50.0	97.7	1.59	654	911
PBTI	32	8	2589 [2370, 2766]	93.2	173.4	1.66	645	911
PBTI	128	1	1908 [1713, 2077]	60.4	140.7	0.96	508	911
PBTI	128	2	2470 [2438, 2492]	98.9	156.8	1.59	642	911
PBTI	128	4	3078 [3039, 3133]	157.5	236.2	1.79	580	911
PBTI	128	8	2989 [2872, 3080]	325.2	497.7	1.78	596	911
PBTI	512	1	1924 [1891, 1948]	259.3	364.8	0.97	504	911
PBTI	512	2	2861 [2732, 2973]	333.9	456.2	1.74	610	911
PBTI	512	4	3184 [3121, 3283]	594.7	879.0	1.81	568	911
PBTI	512	8	2725 [2427, 2894]	1305.2	2067.7	1.70	627	911

Table S38. MQTT v5 QoS 1 over mutual TLS (Mosquitto), publish-to-commit latency, first series (B1–B3 and PBTI; publisher, broker and verifier on one host); means over 5 repetitions, 95% bootstrap CI for p99.

Load (msg/s)	Pipeline	p50 (ms)	p95 (ms)	p99 (ms) [95% CI]	Achieved (msg/s)	Verifier CPU (μs/msg)	Publisher CPU (μs/msg)
200	B1	5.91	11.02	17.40 [16.8, 18.0]	200	691	553
	B2	5.91	11.01	17.22 [16.5, 18.2]	200	706	547
	B3	6.17	11.35	17.19 [16.4, 17.9]	200	939	655
	PBTI	6.27	11.39	14.37 [13.5, 15.4]	200	1123	848
500	B1	6.76	13.74	24.68 [21.7, 29.7]	500	476	442
	B2	6.76	14.01	22.79 [21.9, 23.6]	500	481	439
	B3	6.97	15.13	22.88 [22.3, 23.6]	500	724	543
	PBTI	6.93	12.05	21.98 [20.9, 23.1]	500	920	737
1000	B1	6.84	21.22	40.70 [37.1, 43.6]	1000	405	421
	B2	6.86	21.89	46.04 [39.6, 52.7]	999	415	423
	B3	7.29	22.75	41.91 [37.9, 46.2]	1000	648	497
	PBTI	7.96	29.49	51.20 [46.1, 56.3]	999	788	652
1500	B1	7.53	30.58	79.82 [63.7, 95.9]	1500	352	356
	B2	7.58	30.48	69.67 [62.6, 77.7]	1500	360	362
	B3	10.49	64.28	106.56 [84.1, 130.5]	1499	555	422
	PBTI	660.08	1076.78	1112.37 [838.3, 1386.4]	1413	678	561

Table S39. MQTT v5 QoS 1 over mutual TLS, second series with 30 repetitions per cell (2 s warm-up, 10 s measurement; all roles on one host): latency from publication of the signed message to commit, means over repetitions with 95% bootstrap CIs. At 1000 messages/s the co-located publisher, verifier and broker used about 1.7 of the two vCPUs (769, 879 and 69 μs per message for PBTI) and the median latency of single repetitions ranged from 8.8 to 187.6 ms, which explains the difference from the five-repetition first series (Table S38); the runs locate the saturation of the shared host, not the capacity of a dedicated verifier.

Pipeline	Offered	Reps	Sent msg/s	Committed msg/s	p50 ms [95% CI]	p99 ms [95% CI]	CPU μs/msg	Max queue
B2	1,000	30	1,000	1000	7.2 [7.0, 7.4]	50.8 [46.3, 55.7]	475	53
B2	1,250	30	1250	1249	7.8 [7.6, 7.9]	68.6 [62.1, 75.8]	434	82
B2	1,500	30	1500	1499	8.7 [8.5, 8.8]	89.1 [81.6, 97.3]	402	132
PBTI	1,000	30	999	998	26.2 [16.6, 40.1]	187.9 [140.5, 250.3]	879	195
PBTI	1,250	30	1228	1176	400.7 [268.9, 554.9]	942.3 [724.8, 1181.2]	794	1105
PBTI	1,500	30	1351	1211	1161.2 [983.4, 1349.5]	1819.5 [1624.2, 2034.2]	771	2453

Table S40. SKAB replay through the MQTT/TLS deployment at native rate and 5 and 10× (24 devices at about 1, 5 and 10 rows/s each; achieved rates in the table; 10 repetitions; one host). Lost: sent minus committed.

Speed	Pipeline	msg/s	p50 ms	p99 ms [95% CI]	CPU μs/msg	Max queue	RSS MB	DB B/msg	Lost
1×	B4	26	11.0	36.7 [21.1, 65.2]	2785	20	82	1801	0
1×	B5	26	12.9	25.7 [23.7, 27.9]	3047	22	82	1201	0
1×	PBTI	26	12.6	24.2 [22.2, 27.0]	2983	21	82	1201	0
5×	B4	116	11.3	33.8 [27.0, 44.4]	1141	22	82	1707	0
5×	B5	116	12.5	29.4 [27.8, 31.1]	1314	22	82	1127	0
5×	PBTI	116	12.4	28.6 [27.1, 30.3]	1322	22	82	1126	0
10×	B4	230	11.2	33.5 [31.7, 35.5]	925	22	82	1701	0
10×	B5	230	12.1	32.7 [30.7, 34.6]	1091	22	82	1123	0
10×	PBTI	230	12.1	31.7 [30.0, 33.4]	1065	22	82	1125	0

Table S41. Message-size breakdown of a PBTI message (mean over 2,000 messages encoded by the benchmark's edge encoder; 4 gateways × 6 devices). Bytes are attributed to fields by marginal removal from the canonical JSON; they do not depend on the machine. The CBOR rows are an engineering estimate of the same fields with raw signatures and digests, not an implemented format (both computed by `tools/deploy/message_breakdown.py`). The benchmark runs of Table S35 averaged 641 bytes per message (ratio 2.91) because field values such as sequence numbers grow over their longer runs.

Part of a PBTI message	Bytes
Payload and identifiers also carried by B2 (pl, did, aid, ts, sch)	141
Observation-statement additions (version, type, epoch, sequence, prev digest, key id)	114
Origin signature σ_o (base64url)	96
Relay statement (workload id, key id, odig, relay time, mode, version, type)	175
Relay signature σ_w (base64url)	96
JSON structure	17
Total (B2 message: 220)	639
Estimate in CBOR, text / integer keys (B2: 197 / 147)	499 / 434

Table S42. Deployment readiness: what was measured, where, and what is missing. "One host": a two-vCPU virtual machine (Intel Xeon, 2.1 GHz, 7.8 GB RAM) running every role.

Property	Status	Evidence in this study	Missing for deployment; prepared protocol
Admission and containment correctness	measured in simulation	all scenario stages; 24 emulated devices, virtual time	plant data with recorded incidents
Real sensor dynamics	measured, attacks injected	SKAB replay (Table S32)	recorded plant attacks
In-process verifier throughput	measured, one host	2067 messages/s, 0.19× B2 (Table S35)	more cores, compiled or batch-verifying verifier
Batching and parallel verification	measured, one host	3078 messages/s at batch 128 with four workers, 3184 at batch 512 with a p99 batch latency of 879 ms (Table S37)	more cores
Networked latency (MQTT/TLS)	measured, all roles on one host	p50 26.2 ms and p99 187.9 ms at 1000 messages/s (means of 30 repetitions; per-repetition medians 8.8–187.6 ms); the shared host saturates between 1000 and 1250 messages/s (Table S39)	separate publisher, broker and verifier hosts with measured clock uncertainty (prepared: three-host protocol, `experiments/run_multihost.py`)
Message size	measured exactly	641 versus 220 bytes per message in the benchmark runs (2.91×; Table S35); 639 versus 220 in the field breakdown (Table S41)	binary encoding (estimate only)
Memory growth	measured, in-process	about 2.2 kB of resident memory per message (B2: 0.7 kB); verifier state is never pruned	bounded state (pruning after the epoch drain and horizon)
Storage growth	measured, one host	909–1201 bytes per message persisted depending on the workload (Tables S35, S40), of which evidence about 374	retention and compaction policy
Device-side signing	measured on a server CPU only	291 μs per message	constrained-device benchmark with thermal state and energy (prepared: `experiments/run_device_bench.py`)
Hardware-backed keys	not implemented	–	secure element or TPM support for the signature algorithm, or a COSE algorithm the hardware supports
Clock synchronization	assumed (verifier clock trusted)	–	measured clock-uncertainty bound across hosts
Crash recovery and restarts	not implemented for the verifier (the store is recreated at start-up, acknowledgements precede the commit, the broker runs without persistence); simulated for gateways only	gateway restart during drain (E11/E12); the simulator acknowledges entries end to end	state reload, acknowledgement after commit, persistent sessions; verifier restart, broker outage and interrupted review on real hosts
Concurrent verifiers	not evaluated	workers only pre-verify signatures	protocol for exchanging conflict evidence between replicas
Witness deployment	deterministic matrix only	Table S29	trust and timing model, availability, campaign evaluation
Human review workload	not measured	automated decisions counted	operator study

Table S43. Conceptual alignment of PBTI with standards and reference architectures. This is an architectural mapping, not a claim of compliance or conformance; no certification or conformance testing was performed.

Reference	Alignment
NIST SP 800-207	PBTI applies the decision/enforcement pattern to data objects: the verifier acts as a policy enforcement point and the policy store/PDP as a policy decision point, per data object rather than per session; decisions use identity, dynamic policy and continuously refreshed credential status, and no trust is derived from the network location of the broker.
NIST CSF 2.0	Protect: integrity of data at rest and in transit (signatures, content addressing); Detect: classification of every receipt, gap and conflict evidence; Respond/Recover: revocation, taint propagation and corrected derived versions; Govern/Identify: versioned policy and asset assignment.
IEC 62443-3-3 / 4-2	FR1 identification and authentication control (device and workload credentials); FR2 use control (path-bound admission); FR3 system integrity (signed statements, tamper-evident evidence); FR5 restricted data flow (authorized relay paths); FR6 timely response to events (evidence for incident response, taint); FR4 and FR7 are not addressed.
NISTIR 8259A	Device-side prerequisites: device identification (unique keyed identity; the boot epoch is a PBTI extension, not part of 8259A) and data protection (signing of observations); key storage and update are assumed.
NIST SP 800-82r3	OT data flows from cells to enterprise systems through managed conduits, with the broker treated as untrusted for data integrity.
IIRA v1.10 / RAMI 4.0	Placement in the edge-to-enterprise data path between the edge tier and the platform/enterprise tiers; asset identity follows the plant hierarchy.
W3C PROV	Raw observation → `prov:Entity`; device and workload → `prov:Agent` (relay via `prov:actedOnBehalfOf`); admission and aggregation → `prov:Activity`; derived version `prov:wasDerivedFrom` inputs. Dispositions, contest/taint status, policy version and credential status have no PROV equivalent.
SPIFFE / X.509-SVID	Workload identity in the relay statement and in short-lived workload certificates; SPIRE attestation is a precondition outside the evaluated mechanism.